Block practice · Differentiation and its applications

Work through each question type in order. Each section has a worked-example retry and all further practice. Attempt only the part outlined in red.

Product rule and chain rule: differentiating products and 'function of a function'

The quotient rule: differentiating fractions

Finding stationary points

Maximum or minimum? Using the second derivative

Finding the equation of a tangent

Finding the equation of a normal

Optimisation part 1: building the formula to maximise or minimise

Optimisation part 2: finding, justifying and evaluating the maximum or minimum

Rates of change with one chain-rule link

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Practice workbook · Product rule and chain rule: differentiating products and 'function of a function'

Using the chain rule on composite functions and the product rule on products such as an exponential times a polynomial, a trig function or a root.

10 question parts: 1 worked-example retry and 9 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: FPM-CAL-01

1. Question 6 · part a.i · 3 marks

2019 JUN · Paper 02

Worked-example retry · Attempt part a.i only.

6 (a) Given that $y = (4x - 3)e^{2x}$

(i) find $\frac{dy}{dx}$ (3)

(ii) show that $(4x - 3)\frac{dy}{dx} = (8x - 2)y$ (2)

(b) Differentiate $\frac{\sin 5x}{(x - 3)^2}$ with respect to x (3)

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1. Question 6 · part a.i · 3 marks · Working and answer

Working and answer

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2. Question 1 · part whole · 3 marks

2020 NOV · Paper 01

Further practice · Attempt part whole only.

1 Differentiate with respect to x

$$6e^{3x^2}\cos 2x$$

(3)

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(Total for Question 1 is 3 marks)

2. Question 1 · part whole · 3 marks · Working and answer

Working and answer

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3. Question 11 · part a · 5 marks

2022 JAN · Paper 01

Further practice · Attempt part a only.

11 Given that

$$y = \frac{e^{4x}}{32}(8x^2 - 4x + 1)$$

(a) show that $\frac{dy}{dx} = x^2e^{4x}$

(5)

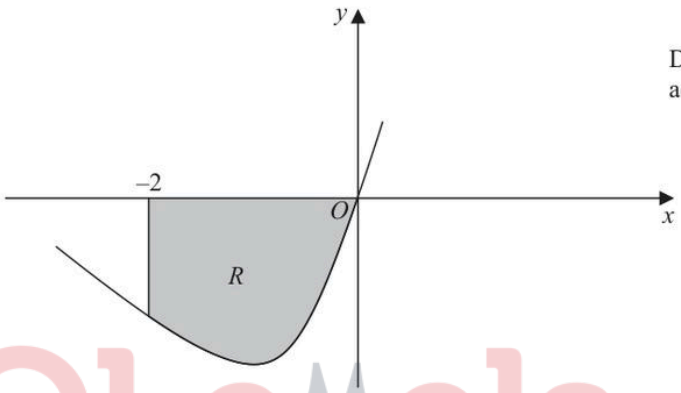


Diagram **NOT** accurately drawn

Figure 2

Figure 2 shows part of the curve C with equation $y = 3xe^{2x}$

The finite region R bounded by C, the straight line with equation $x = -2$ and the x-axis, shown shaded in Figure 2, is rotated through 360° about the x-axis.

(b) Using part (a), find the volume, to 2 significant figures, of the solid formed.

(4)

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3. Question 11 · part a · 5 marks · Working and answer

Working and answer

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4. Question 3 · part a · 3 marks

2022 JAN · Paper 02 (regional R)

Further practice · Attempt part a only.

3 Differentiate with respect to x

(a) $e^{2x}\sqrt{5x-3}$

(3)

(b) $\dfrac{x^3}{\cos 3x}$

(3)

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4. Question 3 · part a · 3 marks · Working and answer

Working and answer

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5. Question 6 · part b · 3 marks

2019 JUN · Paper 02

Further practice · Attempt part b only.

6 (a) Given that $y = (4x - 3)e^{2x}$

(i) find $\frac{dy}{dx}$ (3)

(ii) show that $(4x - 3)\frac{dy}{dx} = (8x - 2)y$ (2)

(b) Differentiate $\frac{\sin 5x}{(x-3)^2}$ with respect to x (3)

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5. Question 6 · part b · 3 marks

2019 JUN · Paper 02

Further practice · Attempt part b only.

Question 6 continued

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(Total for Question 6 is 8 marks)

5. Question 6 · part b · 3 marks · Working and answer

Working and answer

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6. Question 7 · part a · 4 marks

2023 JAN · Paper 02

Further practice · Attempt part a only.

7 $y = e^{2x} \cos 2x$

(a) Show that

$$\frac{dy}{dx} = 2y - 2e^{2x} \sin 2x$$

(4)

(b) Hence show that

$$\frac{d^2y}{dx^2} = 4\frac{dy}{dx} - 8y$$

(5)

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6. Question 7 · part a · 4 marks · Working and answer

Working and answer

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7. Question 9 · part a · 3 marks

2025 JUN · Paper 02

Further practice · Attempt part a only.

9

$$y = e^{-4t} \cos 2t$$

(a) Show that $2e^{-4t}\sin 2t = -\frac{dy}{dt} - 4y$ (3)

Given that $\frac{d^2y}{dt^2} + M\frac{dy}{dt} + Ny = 0$ where M and N are integers

(b) find the value of M and the value of N (5)

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7. Question 9 · part a · 3 marks · Working and answer

Working and answer

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8. Question 3 · part a · 3 marks

2020 JAN · Paper 01 (regional R)

Further practice · Attempt part a only.

3 Given that $y = e^{2x}(x^2 + 1)$

(a) find $\frac{dy}{dx}$

(3)

The straight line l is the tangent to the curve with equation $y = e^{2x}(x^2 + 1)$ at the point on the curve where $x = 0$

(b) Find an equation for l in the form $y = mx + c$

(3)

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8. Question 3 · part a · 3 marks · Working and answer

Working and answer

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9. Question 5 · part a · 5 marks

2022 JUN · Paper 01 (regional R)

Further practice · Attempt part a only.

5 Differentiate with respect to x

(a) $e^{4x}(6x+2)^{\frac{3}{2}}$

Give your answer in the form $e^{4x}\left(\sqrt{6x+2}\right)(Ax+B)$ where A and B are integers.

(5)

(b) $\dfrac{\sin 3x}{(2x-4)^3}$

(3)

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9. Question 5 · part a · 5 marks · Working and answer

Working and answer

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10. Question 9 · part a · 2 marks

2025 JUN · Paper 01 (regional R)

Further practice · Attempt part a only.

9 Given that $f(x) = (1+x^2)^4$

(a) show that $f'(x) = 8x(1+x^2)^3$ (2)

The curve C has equation $y = \dfrac{\sin 3x}{(1+x^2)^4}$

The point A on C has x coordinate $\dfrac{2\pi}{3}$

(b) Show that the gradient of the normal to C at A is

$$-\frac{1}{3}\left(1+\left(\frac{2\pi}{3}\right)^2\right)^4$$ (7)

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10. Question 9 · part a · 2 marks · Working and answer

Working and answer

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Practice workbook · The quotient rule: differentiating fractions

Differentiating a fraction u/v, simplifying the result and using it to give a gradient or to show the gradient always has one sign.

16 question parts: 1 worked-example retry and 15 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: FPM-CAL-02

1. Question 10 · part d · 3 marks

2025 JUN · Paper 02

Worked-example retry · Attempt part d only.

10 A curve C has equation

$$y = \frac{7-x}{2x+3} \qquad x \neq -\frac{3}{2}$$

(a) Write down an equation of the asymptote to C that is

(i) parallel to the y-axis

(ii) parallel to the x-axis

(2)

(b) Find the coordinates of the points of intersection of C with the coordinate axes.

(2)

(c) Using the axes on the opposite page, sketch C, showing clearly the asymptotes and the coordinates of the points where C crosses the coordinate axes.

(3)

C passes through the point A with coordinates $\left(\frac{1}{2}, \frac{13}{8}\right)$

(d) Show that the gradient of C at A is $-\frac{17}{16}$

(3)

C also passes through the point P

Given that the tangent to C at P is parallel to the tangent to C at A

(e) find an equation of the tangent to C at P
Give your answer in the form $ax + by + c = 0$ where a, b and c are integers.

(7)

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1. Question 10 · part d · 3 marks · Working and answer

Working and answer

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2. Question 8 · part a · 5 marks

2020 NOV · Paper 01

Further practice · Attempt part a only.

8 Given that $2xy + 5y = e^x$

(a) show that $\frac{dy}{dx} = \frac{y(2x+3)}{2x+5}$

(5)

(b) find the value of $\frac{dy}{dx}$ when $x = 0$

(2)

(c) find an equation of the normal to the curve with equation $2xy + 5y = e^x$ at the point where $x = 0$
Give your answer in the form $px + qy + r = 0$ where p, q and r are integers.

(3)

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2. Question 8 · part a · 5 marks · Working and answer

Working and answer

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3. Question 8 · part b · 2 marks

2020 NOV · Paper 01

Further practice · Attempt part b only.

8 Given that $2xy + 5y = e^x$

(a) show that $\frac{dy}{dx} = \frac{y(2x+3)}{2x+5}$

(5)

(b) find the value of $\frac{dy}{dx}$ when $x = 0$

(2)

(c) find an equation of the normal to the curve with equation $2xy + 5y = e^x$ at the point where $x = 0$
Give your answer in the form $px + qy + r = 0$ where p, q and r are integers.

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3. Question 8 · part b · 2 marks · Working and answer

Working and answer

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4. Question 3 · part b · 3 marks

2022 JAN · Paper 02 (regional R)

Further practice · Attempt part b only.

3 Differentiate with respect to x

(a) $e^{2x}\sqrt{5x-3}$

(3)

(b) $\dfrac{x^3}{\cos 3x}$

(3)

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4. Question 3 · part b · 3 marks

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Further practice · Attempt part b only.

Question 3 continued

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(Total for Question 3 is 6 marks)

4. Question 3 · part b · 3 marks · Working and answer

Working and answer

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5. Question 8 · part whole · 5 marks

2025 JUN · Paper 02 (regional R)

Further practice · Attempt part whole only.

8 Use calculus to find the gradient of the curve with equation

$$y = e^{3x}(9x+2)^{\frac{1}{3}}$$

at the point on the curve where $x = 1$
Give your answer to 3 significant figures.

(5)

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5. Question 8 · part whole · 5 marks

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Further practice · Attempt part whole only.

Question 8 continued

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(Total for Question 8 is 5 marks)

5. Question 8 · part whole · 5 marks · Working and answer

Working and answer

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6. Question 10 · part d · 4 marks

2025 JUN · Paper 02 (regional R)

Further practice · Attempt part d only.

10 The curve C has equation

$$y = \frac{5x-2}{ax+b} \qquad x \neq -\frac{b}{a}$$

The asymptote to C that is parallel to the y-axis has equation $x = -\frac{4}{3}$

C crosses the y-axis at the point with coordinates $\left(0, -\frac{1}{4}\right)$

(a) (i) Show that $b = 8$

(ii) Find the value of a

(3)

(b) Write down the equation of the asymptote to C that is parallel to the x-axis.

(1)

(c) Find the coordinates of the point where C crosses the x-axis.

(1)

(d) Using calculus, show that at every point on C, the gradient is positive.

(4)

(e) Using the axes on the next page, sketch C
Clearly label the asymptotes and the coordinates where C crosses the coordinate axes.

(3)

The gradient of C is $\frac{13}{100}$ at the point E and at the point F

(f) Find the length of the line EF, giving your answer to 2 decimal places.

(6)

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6. Question 10 · part d · 4 marks · Working and answer

Working and answer

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7. Question 7 · part b · 4 marks

2021 JUN · Paper 01

Further practice · Attempt part b only.

7 The curve C has equation $y = \dfrac{x}{x^2 + 4}$

(a) Using calculus, find the coordinates of the stationary points on C.

(5)

(b) Show that $\dfrac{d^2y}{dx^2} = \dfrac{2x(x^2 - 12)}{(x^2 + 4)^3}$

(4)

(c) Hence, or otherwise, determine the nature of each of these stationary points.

(2)

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7. Question 7 · part b · 4 marks · Working and answer

Working and answer

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8. Question 5 · part b · 3 marks

2022 JUN · Paper 01 (regional R)

Further practice · Attempt part b only.

5 Differentiate with respect to x

(a) $e^{4x}(6x+2)^{\frac{3}{2}}$

Give your answer in the form $e^{4x}\left(\sqrt{6x+2}\right)(Ax+B)$ where A and B are integers.

(5)

(b) $\dfrac{\sin 3x}{(2x-4)^3}$

(3)

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8. Question 5 · part b · 3 marks

2022 JUN · Paper 01 (regional R)

Further practice · Attempt part b only.

Question 5 continued

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(Total for Question 5 is 8 marks)

8. Question 5 · part b · 3 marks · Working and answer

Working and answer

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9. Question 2 · part whole · 5 marks

2023 JUN · Paper 01

Further practice · Attempt part whole only.

2 $y = (\sin 2x)\sqrt{3 + 2x}$

Show that $\frac{dy}{dx} = \frac{\sin 2x + (A + Bx)\cos 2x}{\sqrt{3 + 2x}}$ where A and B are integers to be found. **(5)**

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9. Question 2 · part whole · 5 marks

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Question 2 continued

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(Total for Question 2 is 5 marks)

9. Question 2 · part whole · 5 marks · Working and answer

Working and answer

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10. Question 4 · part whole · 5 marks

2021 NOV · Paper 01

Further practice · Attempt part whole only.

4 $y = \dfrac{\sin 2x}{\sqrt{x^2 - 9}} \qquad |x| > 3$

Show that $\dfrac{dy}{dx} = \dfrac{2(x^2 - 9)\cos 2x - x\sin 2x}{\sqrt{(x^2 - 9)^3}}$

(5)

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10. Question 4 · part whole · 5 marks

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Further practice · Attempt part whole only.

Question 4 continued

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(Total for Question 4 is 5 marks)

10. Question 4 · part whole · 5 marks · Working and answer

Working and answer

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11. Question 10 · part c · 4 marks

2022 JUN · Paper 02 (regional R)

Further practice · Attempt part c only.

10 A curve C has equation

$$y = \frac{7x - 2}{2x - 3} \qquad x \neq \frac{3}{2}$$

(a) Write down an equation of the asymptote to C that is

(i) parallel to the y-axis,

(ii) parallel to the x-axis.

(2)

(b) Find the coordinates of the points of intersection of C with the coordinate axes.

(2)

(c) Using calculus, show that at every point on the curve, the gradient of C is negative.

(4)

(d) Using the axes on the opposite page, sketch C.
Show clearly and label with their equation any asymptotes and the coordinates of the points of intersection of C with the coordinate axes.

(3)

The straight line l is the normal to C at the point A.
The x coordinate of A is positive and the gradient of l is 17
The line l also intersects C at the point B.

(e) Find the exact coordinates of B.

(7)

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11. Question 10 · part c · 4 marks · Working and answer

Working and answer

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12. Question 9 · part d · 3 marks

2023 JUN · Paper 02 (regional R)

Further practice · Attempt part d only.

9 A curve C has equation $y = \dfrac{3 - 2x}{x + 6}$ where $x \neq -6$

(a) Write down an equation of the asymptote to C that is parallel to the

(i) x-axis (ii) y-axis

(2)

(b) Find the coordinates of the point where C crosses the

(i) x-axis (ii) y-axis

(2)

(c) Using the axes opposite, sketch the graph of C, showing clearly its asymptotes and the coordinates of the points where C crosses the coordinate axes.

(3)

(d) Show that the gradient of the tangent to C is always negative.

(3)

A tangent to C has equation $y = -\dfrac{3}{5}x + k$ where $k > 0$

(e) Find the value of k

(5)

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12. Question 9 · part d · 3 marks · Working and answer

Working and answer

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13. Question 11 · part a · 3 marks

2022 JUN · Paper 02

Further practice · Attempt part a only.

11 A curve C has equation

$$y = \frac{(2a-1)x+1}{ax-6} \qquad \text{where } a \text{ is a constant and } x \neq \frac{6}{a}$$

(a) Find $\frac{dy}{dx}$

(3)

The curve crosses the y-axis at the point A.

The normal to C at the point A is the line l with equation $66y - 72x + 11 = 0$

Show that

(b) (i) $a = 3$

(4)

(ii) the equation of C is $y = \frac{5x+1}{3x-6}$ where $x \neq 2$

(1)

(c) Using the axes on the opposite page, sketch C, showing clearly the asymptotes with their equations and the coordinates of the points where C crosses the coordinate axes.

(5)

The line l meets C again at the point D.

(d) Find the x coordinate of D.
Give your answer as an improper fraction.

(4)

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13. Question 11 · part a · 3 marks · Working and answer

Working and answer

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14. Question 3 · part whole · 5 marks

2024 NOV · Paper 02

Further practice · Attempt part whole only.

3 The curve C has equation $y = e^{3x}(2x-1)^4$

Using calculus, find the exact value of the gradient of the tangent to C when $x = 1$

(5)

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14. Question 3 · part whole · 5 marks

2024 NOV · Paper 02

Further practice · Attempt part whole only.

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Question 3 continued

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(Total for Question 3 is 5 marks)

14. Question 3 · part whole · 5 marks · Working and answer

Working and answer

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15. Question 6 · part a · 5 marks

2023 JAN · Paper 01 (regional R)

Further practice · Attempt part a only.

6 The curve C has equation $y = \dfrac{e^{(x^2+1)}}{x^2+1}$

(a) Show that $\dfrac{dy}{dx} = \dfrac{Kx^3e^{(x^2+1)}}{(x^2+1)^2}$ where K is a constant whose value is to be found.

(5)

(b) Find an equation of the tangent to C at the point on C where $x = -1$
Simplify your answer.

(5)

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15. Question 6 · part a · 5 marks · Working and answer

Working and answer

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16. Question 8 · part a · 5 marks

2023 NOV · Paper 01

Further practice · Attempt part a only.

8 $y = \dfrac{2e^{3x+1}}{5x^2}$

(a) Find $\dfrac{dy}{dx}$

Give your answer in the form $\dfrac{Ae^{3x+1}(Bx-A)}{Cx^3}$ where A, B and C are prime numbers to be found.

(5)

The value of x increases by 2%

(b) Use your answer to part (a) to find an estimate, in terms of x, for the percentage change in y

Give your answer in the form $(Px-Q)$ where P and Q are integers.

(3)

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16. Question 8 · part a · 5 marks · Working and answer

Working and answer

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Practice workbook · Finding stationary points

Setting dy/dx = 0 and solving (quadratics, factorised cubics or trig equations) to find the coordinates of stationary points.

8 question parts: 1 worked-example retry and 7 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: FPM-CAL-08

1. Question 7 · part a · 5 marks

2021 JUN · Paper 01

Worked-example retry · Attempt part a only.

7 The curve C has equation $y = \dfrac{x}{x^2 + 4}$

(a) Using calculus, find the coordinates of the stationary points on C.

(5)

(b) Show that $\dfrac{d^2y}{dx^2} = \dfrac{2x(x^2 - 12)}{(x^2 + 4)^3}$

(4)

(c) Hence, or otherwise, determine the nature of each of these stationary points.

(2)

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1. Question 7 · part a · 5 marks · Working and answer

Working and answer

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2. Question 9 · part b.ii · 7 marks

2019 JUN · Paper 02

Further practice · Attempt part b.ii only.

9 The curve C, with equation $y = f(x)$, passes through the point with coordinates $\left(-2, -\frac{28}{3}\right)$

Given that $f'(x) = x^3 - x^2 - 4x + 4$

(a) show that C passes through the origin.

(4)

(b) (i) Show that C has a minimum point at $x = 2$ and a maximum point at $x = 1$

(ii) Find the exact value of the y coordinate at each of these points.

(7)

The curve has another turning point at A.

(c) (i) Find the coordinates of A.

(ii) Determine the nature of this turning point.

(3)

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OLevels.com SCHOOL

2. Question 9 · part b.ii · 7 marks · Working and answer

Working and answer

OLevels.com SCHOOL

3. Question 9 · part c.i · 0 marks

2019 JUN · Paper 02

Further practice · Attempt part c.i only.

9 The curve C, with equation $y = f(x)$, passes through the point with coordinates $\left(-2, -\frac{28}{3}\right)$

Given that $f'(x) = x^3 - x^2 - 4x + 4$

(a) show that C passes through the origin.

(4)

(b) (i) Show that C has a minimum point at $x = 2$ and a maximum point at $x = 1$

(ii) Find the exact value of the y coordinate at each of these points.

(7)

The curve has another turning point at A.

(c) (i) Find the coordinates of A.

(ii) Determine the nature of this turning point.

(3)

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3. Question 9 · part c.i · 0 marks · Working and answer

Working and answer

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4. Question 9 · part c · 6 marks

2020 NOV · Paper 02

Further practice · Attempt part c only.

9 A curve C has equation

$$y = \frac{2 + 4x - x^2}{2x + 1} \quad x \neq -\frac{1}{2}$$

(a) Write the equation of C in the form $ax^2 + (by - 4)x + (y - c) = 0$, where a, b and c are integers whose values are to be found.

(3)

(b) Hence show that x is real when $y \leqslant 2$ and when $y \geqslant 3$

(4)

(c) Find the coordinates of the stationary points on C.

(6)

(d) Sketch C showing clearly

(i) the exact coordinates of the points where C crosses the x-axis,

(ii) the asymptote to C that is parallel to the y-axis,

(iii) the coordinates of the stationary points.

(5)

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P 6 2 2 8 5 A 0 2 4 3 2

4. Question 9 · part c · 6 marks · Working and answer

Working and answer

OLevels.com SCHOOL

5. Question 11 · part c · 7 marks

2024 JUN · Paper 02 (regional R)

Further practice · Attempt part c only.

11 (a) Using a formula on page 2 show that $\cos 2A = 2\cos^2 A - 1$

(2)

(b) Hence show that $\left(2\cos^2 A - 1\right)^2 = \dfrac{\cos 4A + 1}{2}$

(3)

The curve with equation $y = \dfrac{\sin 2x}{2} + \dfrac{\left(2\cos^2 x - 1\right)^2}{2} + \dfrac{1}{8}$ has a stationary point P in the range $0 \leqslant x \leqslant \dfrac{\pi}{6}$

(c) Find the exact coordinates of P

(7)

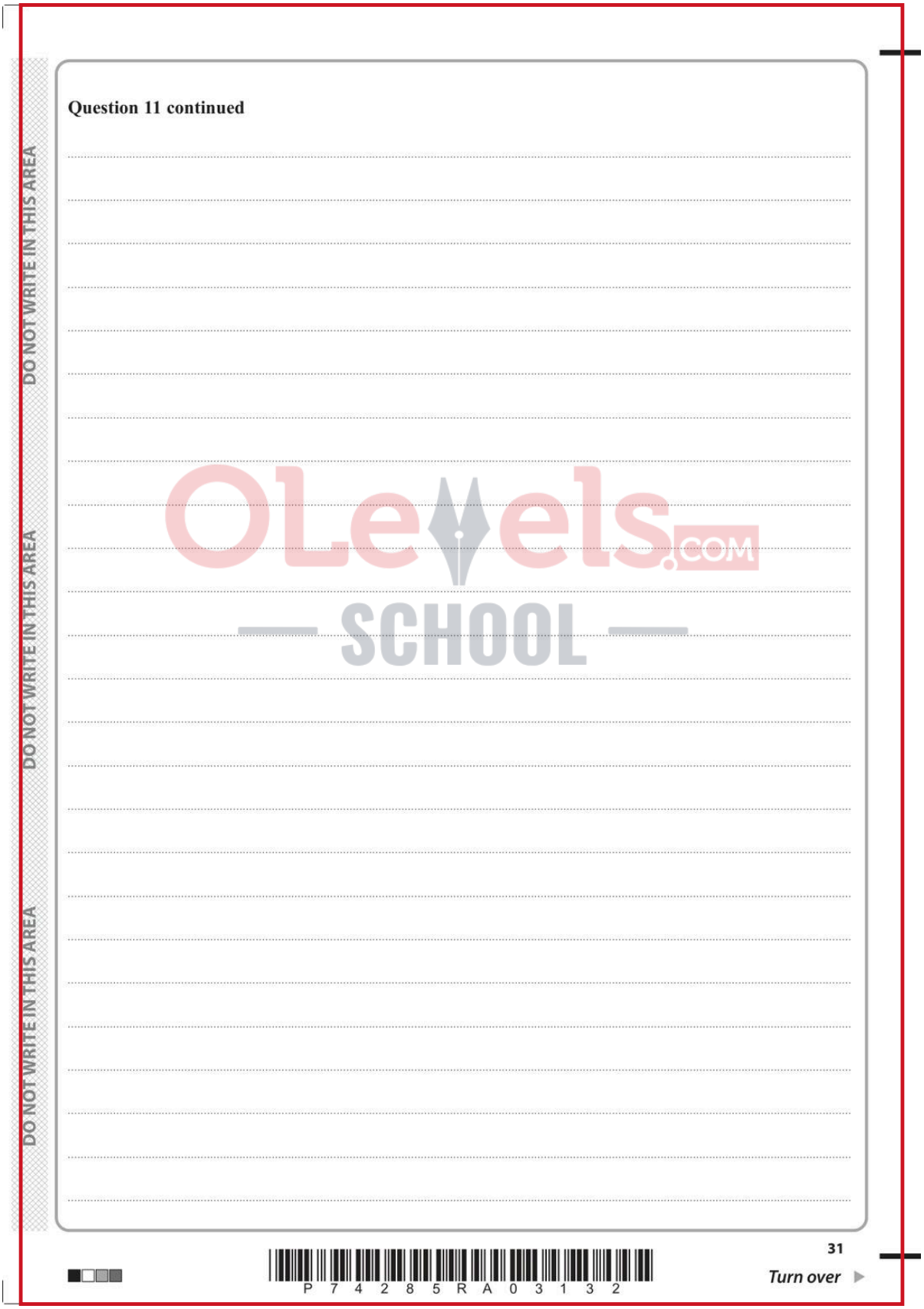
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5. Question 11 · part c · 7 marks

2024 JUN · Paper 02 (regional R)

Further practice · Attempt part c only.



5. Question 11 · part c · 7 marks

2024 JUN · Paper 02 (regional R)

Further practice · Attempt part c only.

Question 11 continued

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(Total for Question 11 is 12 marks)

5. Question 11 · part c · 7 marks · Working and answer

Working and answer

OLevels.com SCHOOL

6. Question 7 · part b.i · 5 marks

2022 JUN · Paper 01

Further practice · Attempt part b.i only.

7 The point with coordinates $(4, -104)$ lies on the curve C with equation $y = f(x)$

Given that $f'(x) = 4x^3 - 12x^2 - 19x + 12$

(a) (i) show that C passes through the origin,

(4)

(ii) show that C has a maximum at the point on the curve where $x = 0.5$

(3)

The curve C has another turning point at A and another turning point at B.

Given that the x coordinate of A is negative,

(b) (i) find the coordinates of A and the coordinates of B,

(5)

(ii) determine the nature of these turning points.

(3)

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P 7 1 6 6 1 A 0 2 0 3 2

6. Question 7 · part b.i · 5 marks · Working and answer

Working and answer

OLevels.com SCHOOL

7. Question 10 · part b · 4 marks

2021 NOV · Paper 01

Further practice · Attempt part b only.

10

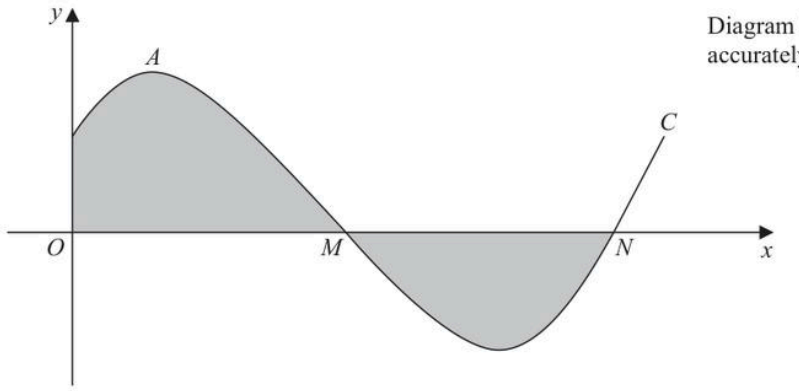


Diagram **NOT** accurately drawn

Figure 4

Figure 4 shows the curve C with equation $y = \frac{1}{2} + \sin 3x$ where $0 \leqslant x < \frac{2\pi}{3}$

The curve C crosses the x-axis at the points M and N

(a) Show that the coordinates of M are $\left(\frac{7\pi}{18}, 0\right)$ and find the coordinates of N

(3)

The curve C has a maximum at the point A

(b) Find the coordinates of A

(4)

(c) Find an equation of the tangent to C at M

Give your answer in the form $ay + b\sqrt{3}x - c\sqrt{3}\pi = 0$ where a, b and c are integers to be found.

(4)

The finite region, shown shaded in Figure 4, is bounded by the curve C, the y-axis and the part of the x-axis from O to N

(d) Use algebraic integration to find, to 3 significant figures, the total area of the shaded region.

(4)

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7. Question 10 · part b · 4 marks · Working and answer

Working and answer

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8. Question 4 · part a · 4 marks

2020 NOV · Paper 01 (regional R)

Further practice · Attempt part a only.

4 The curve C has equation

$$y = x^3 - 3x^2 - 24x + 6$$

(a) Use calculus to find the coordinates of each of the stationary points on C.

(4)

(b) Determine the nature of each of these stationary points.
Justify your answers.

(2)

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8. Question 4 · part a · 4 marks · Working and answer

Working and answer

OLevels.com SCHOOL

Practice workbook · Maximum or minimum? Using the second derivative

Showing a point is stationary and deciding whether it is a maximum or minimum from the sign of d2y/dx2.

8 question parts: 1 worked-example retry and 7 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: FPM-CAL-09

1. Question 7 · part c · 2 marks

2021 JUN · Paper 01

Worked-example retry · Attempt part c only.

7 The curve C has equation $y = \dfrac{x}{x^2 + 4}$

(a) Using calculus, find the coordinates of the stationary points on C.

(5)

(b) Show that $\dfrac{d^2y}{dx^2} = \dfrac{2x(x^2 - 12)}{(x^2 + 4)^3}$

(4)

(c) Hence, or otherwise, determine the nature of each of these stationary points.

(2)

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1. Question 7 · part c · 2 marks

2021 JUN · Paper 01

Worked-example retry · Attempt part c only.

Question 7 continued

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1. Question 7 · part c · 2 marks

2021 JUN · Paper 01

Worked-example retry · Attempt part c only.

Question 7 continued

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1. Question 7 · part c · 2 marks

2021 JUN · Paper 01

Worked-example retry · Attempt part c only.

Question 7 continued

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(Total for Question 7 is 11 marks)

1. Question 7 · part c · 2 marks · Working and answer

Working and answer

OLevels.com SCHOOL

2. Question 9 · part b.i · 0 marks

2019 JUN · Paper 02

Further practice · Attempt part b.i only.

9 The curve C, with equation $y = f(x)$, passes through the point with coordinates $\left(-2, -\frac{28}{3}\right)$

Given that $f'(x) = x^3 - x^2 - 4x + 4$

(a) show that C passes through the origin.

(4)

(b) (i) Show that C has a minimum point at $x = 2$ and a maximum point at $x = 1$

(ii) Find the exact value of the y coordinate at each of these points.

(7)

The curve has another turning point at A.

(c) (i) Find the coordinates of A.

(ii) Determine the nature of this turning point.

(3)

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2. Question 9 · part b.i · 0 marks · Working and answer

Working and answer

OLevels.com SCHOOL

3. Question 9 · part c.ii · 3 marks

2019 JUN · Paper 02

Further practice · Attempt part c.ii only.

9 The curve C, with equation $y = f(x)$, passes through the point with coordinates $\left(-2, -\frac{28}{3}\right)$

Given that $f'(x) = x^3 - x^2 - 4x + 4$

(a) show that C passes through the origin.

(4)

(b) (i) Show that C has a minimum point at $x = 2$ and a maximum point at $x = 1$

(ii) Find the exact value of the y coordinate at each of these points.

(7)

The curve has another turning point at A.

(c) (i) Find the coordinates of A.

(ii) Determine the nature of this turning point.

(3)

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3. Question 9 · part c.ii · 3 marks

2019 JUN · Paper 02

Further practice · Attempt part c.ii only.

Question 9 continued

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3. Question 9 · part c.ii · 3 marks

2019 JUN · Paper 02

Further practice · Attempt part c.ii only.

Question 9 continued

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3. Question 9 · part c.ii · 3 marks

2019 JUN · Paper 02

Further practice · Attempt part c.ii only.

Question 9 continued

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(Total for Question 9 is 14 marks)

3. Question 9 · part c.ii · 3 marks · Working and answer

Working and answer

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4. Question 7 · part b · 2 marks

2023 JUN · Paper 01 (regional R)

Further practice · Attempt part b only.

7 The curve with equation $y = mx^2 + 64\sqrt{x} + 39$ has a stationary point with coordinates $(4, n)$ where m and n are integers.

Using calculus

(a) find the value of m and the value of n

(6)

(b) determine the nature of the stationary point.

(2)

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4. Question 7 · part b · 2 marks

2023 JUN · Paper 01 (regional R)

Further practice · Attempt part b only.

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Question 7 continued

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(Total for Question 7 is 8 marks)

4. Question 7 · part b · 2 marks · Working and answer

Working and answer

OLevels.com SCHOOL

5. Question 7 · part a.ii · 3 marks

2022 JUN · Paper 01

Further practice · Attempt part a.ii only.

7 The point with coordinates $(4, -104)$ lies on the curve C with equation $y = f(x)$

Given that $f'(x) = 4x^3 - 12x^2 - 19x + 12$

(a) (i) show that C passes through the origin,

(4)

(ii) show that C has a maximum at the point on the curve where $x = 0.5$

(3)

The curve C has another turning point at A and another turning point at B.

Given that the x coordinate of A is negative,

(b) (i) find the coordinates of A and the coordinates of B,

(5)

(ii) determine the nature of these turning points.

(3)

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P 7 1 6 6 1 A 0 2 0 3 2

5. Question 7 · part a.ii · 3 marks · Working and answer

Working and answer

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6. Question 7 · part b.ii · 3 marks

2022 JUN · Paper 01

Further practice · Attempt part b.ii only.

7 The point with coordinates $(4, -104)$ lies on the curve C with equation $y = f(x)$

Given that $f'(x) = 4x^3 - 12x^2 - 19x + 12$

(a) (i) show that C passes through the origin,

(4)

(ii) show that C has a maximum at the point on the curve where $x = 0.5$

(3)

The curve C has another turning point at A and another turning point at B.

Given that the x coordinate of A is negative,

(b) (i) find the coordinates of A and the coordinates of B,

(5)

(ii) determine the nature of these turning points.

(3)

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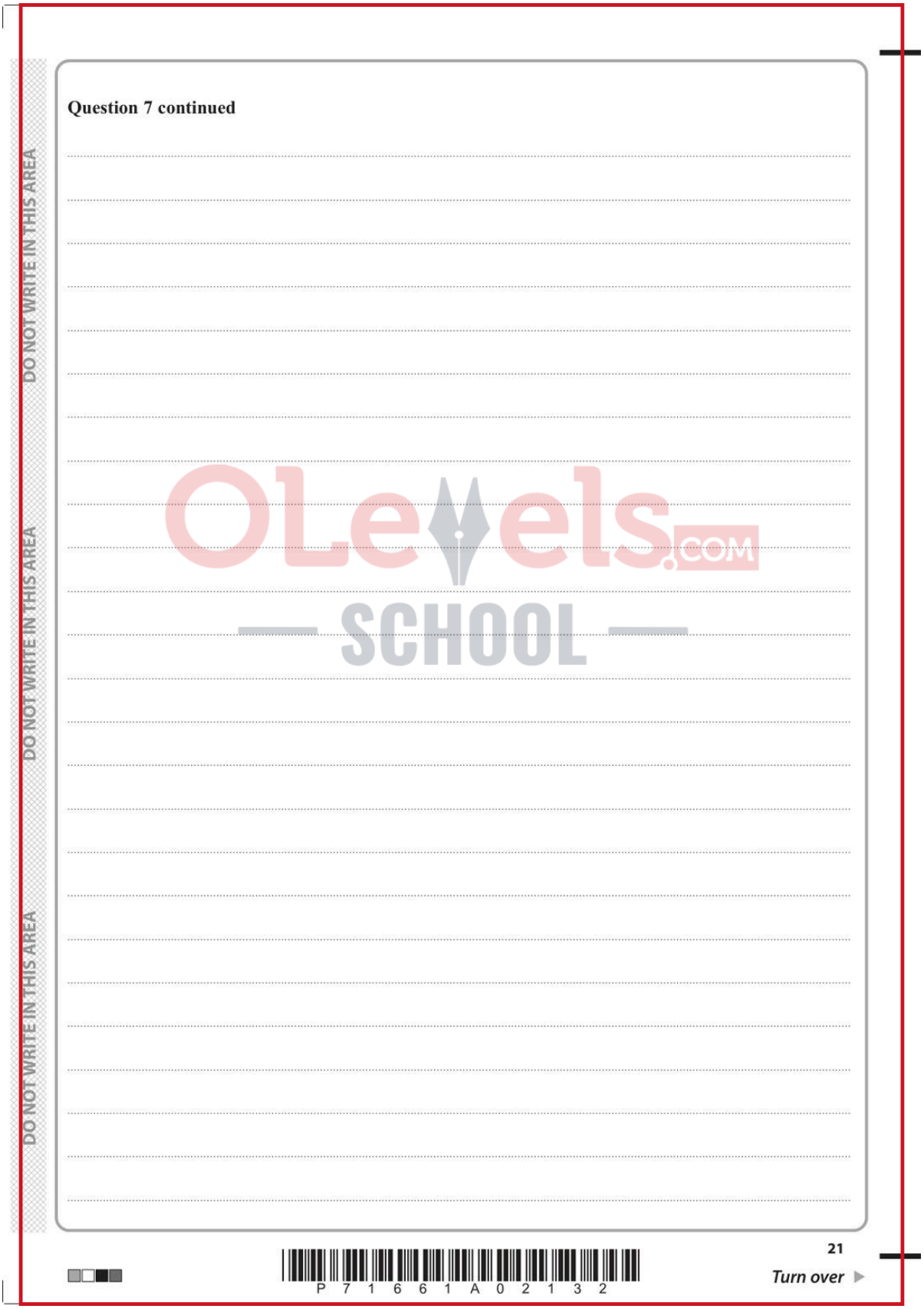
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6. Question 7 · part b.ii · 3 marks

2022 JUN · Paper 01

Further practice · Attempt part b.ii only.



6. Question 7 · part b.ii · 3 marks

2022 JUN · Paper 01

Further practice · Attempt part b.ii only.

Question 7 continued

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6. Question 7 · part b.ii · 3 marks

2022 JUN · Paper 01

Further practice · Attempt part b.ii only.



Question 7 continued

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(Total for Question 7 is 15 marks)

6. Question 7 · part b.ii · 3 marks · Working and answer

Working and answer

OLevels.com SCHOOL

7. Question 4 · part b · 1 mark

2021 NOV · Paper 02

Further practice · Attempt part b only.

4 The curve C has equation $y = 2x^2 + px + q$ where p and q are integers.

The curve C has a stationary point at $(3, -5)$

(a) Show that $p = -12$ and find the value of q

(4)

(b) State, giving a reason, the nature of the stationary point.

(1)

(c) Find an equation of the normal to C at the point on C where $x = 1$

Give your answer in the form $ax + by + c = 0$

(6)

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7. Question 4 · part b · 1 mark · Working and answer

Working and answer

OLevels.com SCHOOL

8. Question 4 · part b · 2 marks

2020 NOV · Paper 01 (regional R)

Further practice · Attempt part b only.

4 The curve C has equation

$$y = x^3 - 3x^2 - 24x + 6$$

(a) Use calculus to find the coordinates of each of the stationary points on C.

(4)

(b) Determine the nature of each of these stationary points.
Justify your answers.

(2)

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8. Question 4 · part b · 2 marks

2020 NOV · Paper 01 (regional R)

Further practice · Attempt part b only.

Question 4 continued

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(Total for Question 4 is 6 marks)

8. Question 4 · part b · 2 marks · Working and answer

Working and answer

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Practice workbook · Finding the equation of a tangent

Finding the gradient at a point by differentiating and writing the tangent's equation, sometimes then finding where it meets an axis or another line.

9 question parts: 1 worked-example retry and 8 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

OLevels.com SCHOOL

Type ID: FPM-CAL-04

1. Question 8 · part b · 5 marks

2020 NOV · Paper 02

Worked-example retry · Attempt part b only.

8 The curve C_1 has equation $y = 5e^{-2x} + 4$

The curve C_2 has equation $y = e^{2x}$

The curves C_1 and C_2 intersect at the point A.

(a) Find the exact coordinates of A.

(4)

The tangent at A to C_1 intersects the x-axis at the point B.

(b) Show that the x coordinate of B is $\frac{1}{2}(5 + \ln 5)$

(5)

The tangent at A to C_2 intersects the x-axis at the point D.

(c) Find the area of ΔABD.

(6)

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P 6 2 2 8 5 A 0 2 0 3 2

1. Question 8 · part b · 5 marks · Working and answer

Working and answer

OLevels.com SCHOOL

2. Question 6 · part b · 6 marks

2020 NOV · Paper 02 (regional R)

Further practice · Attempt part b only.

6

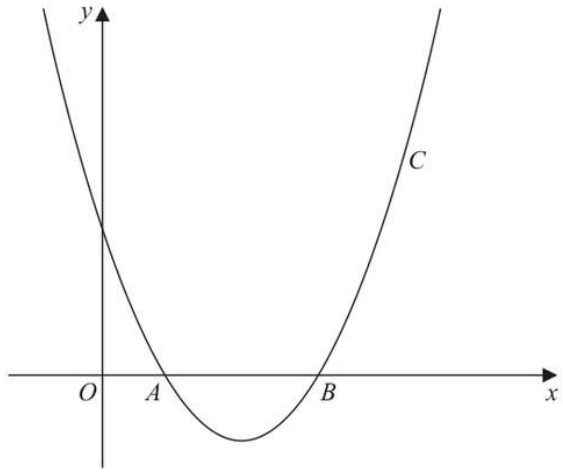


Diagram **NOT** accurately drawn

Figure 3

The curve C with equation $y = x^2 - 5x + 4$ crosses the x-axis at the points A and B, as shown in Figure 3

(a) Find the coordinates of A and the coordinates of B.

(3)

The tangent to C at A meets the tangent to C at B at the point T.

(b) Find the coordinates of T.

(6)

The normal to C at A meets the normal to C at B at the point N.

(c) Find the coordinates of N.

(3)

(d) Find the area of the quadrilateral $ATBN$.

(3)

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2. Question 6 · part b · 6 marks · Working and answer

Working and answer

OLevels.com SCHOOL

3. Question 6 · part a · 5 marks

2022 JAN · Paper 02 (regional R)

Further practice · Attempt part a only.

6 The curve C has equation

$$y = 4\sqrt{x}$$

The point A on C has coordinates (9, 12)

The tangent to C at the point A meets the x-axis at the point T

(a) Find the coordinates of T

(5)

The normal to C at the point A meets the x-axis at the point N

(b) Find the coordinates of N

(4)

(c) Calculate the area of triangle ATN

(2)

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3. Question 6 · part a · 5 marks · Working and answer

Working and answer

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4. Question 8 · part c · 6 marks

2020 NOV · Paper 02

Further practice · Attempt part c only.

8 The curve C_1 has equation $y = 5e^{-2x} + 4$

The curve C_2 has equation $y = e^{2x}$

The curves C_1 and C_2 intersect at the point A.

(a) Find the exact coordinates of A.

(4)

The tangent at A to C_1 intersects the x-axis at the point B.

(b) Show that the x coordinate of B is $\frac{1}{2}(5 + \ln 5)$

(5)

The tangent at A to C_2 intersects the x-axis at the point D.

(c) Find the area of ΔABD.

(6)

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4. Question 8 · part c · 6 marks

2020 NOV · Paper 02

Further practice · Attempt part c only.

Question 8 continued

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4. Question 8 · part c · 6 marks

2020 NOV · Paper 02

Further practice · Attempt part c only.

Question 8 continued

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4. Question 8 · part c · 6 marks

2020 NOV · Paper 02

Further practice · Attempt part c only.

Question 8 continued

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(Total for Question 8 is 15 marks)

4. Question 8 · part c · 6 marks · Working and answer

Working and answer

OLevels.com SCHOOL

5. Question 4 · part whole · 7 marks

2023 JAN · Paper 02

Further practice · Attempt part whole only.

4 The equation of a curve is $y = x^3 \sin x$

Find an equation of the tangent to the curve at the point on the curve where $x = \frac{1}{2}\pi$

Give your answer in the form $y = mx + c$

(7)

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5. Question 4 · part whole · 7 marks

2023 JAN · Paper 02

Further practice · Attempt part whole only.

Question 4 continued

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(Total for Question 4 is 7 marks)

5. Question 4 · part whole · 7 marks · Working and answer

Working and answer

OLevels.com SCHOOL

6. Question 3 · part b · 3 marks

2020 JAN · Paper 01 (regional R)

Further practice · Attempt part b only.

3 Given that $y = e^{2x}(x^2 + 1)$

(a) find $\frac{dy}{dx}$

(3)

The straight line l is the tangent to the curve with equation $y = e^{2x}(x^2 + 1)$ at the point on the curve where $x = 0$

(b) Find an equation for l in the form $y = mx + c$

(3)

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P 6 0 4 7 5 A 0 6 3 6

6. Question 3 · part b · 3 marks

2020 JAN · Paper 01 (regional R)

Further practice · Attempt part b only.

Question 3 continued

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(Total for Question 3 is 6 marks)

6. Question 3 · part b · 3 marks · Working and answer

Working and answer

OLevels.com SCHOOL

7. Question 10 · part a · 5 marks

2025 NOV · Paper 02

Further practice · Attempt part a only.

10

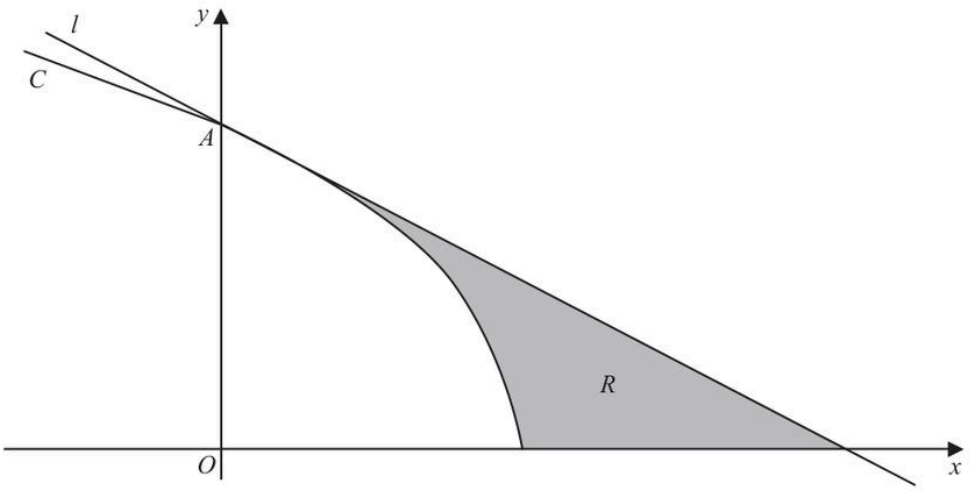


Figure 3

Figure 3 shows part of the curve C with equation $y = 3\sqrt{4-x}$ and part of the line l with equation $px + qy + r = 0$ where p and r are integers and q is prime.

The curve C cuts the y-axis at the point A

The line l is the tangent to C at A

(a) Find the value of p, the value of q and the value of r

(5)

The finite region R, shown shaded in Figure 3, is rotated through 360° about the x-axis.

(b) Use algebraic integration to find the volume of the solid formed.
Give your answer in terms of π

(5)

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7. Question 10 · part a · 5 marks · Working and answer

Working and answer

OLevels.com SCHOOL

8. Question 10 · part c · 4 marks

2021 NOV · Paper 01

Further practice · Attempt part c only.

10

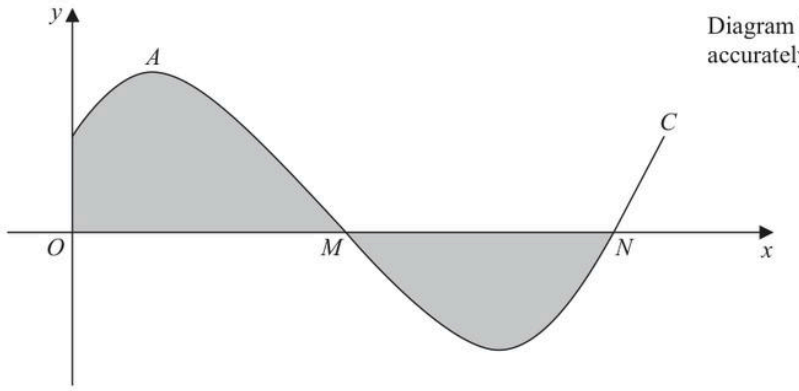


Diagram **NOT** accurately drawn

Figure 4

Figure 4 shows the curve C with equation $y = \frac{1}{2} + \sin 3x$ where $0 \leqslant x < \frac{2\pi}{3}$

The curve C crosses the x-axis at the points M and N

(a) Show that the coordinates of M are $\left(\frac{7\pi}{18}, 0\right)$ and find the coordinates of N

(3)

The curve C has a maximum at the point A

(b) Find the coordinates of A

(4)

(c) Find an equation of the tangent to C at M

Give your answer in the form $ay + b\sqrt{3}x - c\sqrt{3}\pi = 0$ where a, b and c are integers to be found.

(4)

The finite region, shown shaded in Figure 4, is bounded by the curve C, the y-axis and the part of the x-axis from O to N

(d) Use algebraic integration to find, to 3 significant figures, the total area of the shaded region.

(4)

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8. Question 10 · part c · 4 marks · Working and answer

Working and answer



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9. Question 6 · part b · 5 marks

2023 JAN · Paper 01 (regional R)

Further practice · Attempt part b only.

6 The curve C has equation $y = \dfrac{e^{(x^2+1)}}{x^2+1}$

(a) Show that $\dfrac{dy}{dx} = \dfrac{Kx^3e^{(x^2+1)}}{(x^2+1)^2}$ where K is a constant whose value is to be found.

(5)

(b) Find an equation of the tangent to C at the point on C where $x = -1$
Simplify your answer.

(5)

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9. Question 6 · part b · 5 marks

2023 JAN · Paper 01 (regional R)

Further practice · Attempt part b only.

Question 6 continued

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(Total for Question 6 is 10 marks)

9. Question 6 · part b · 5 marks · Working and answer

Working and answer

OLevels.com SCHOOL

Practice workbook · Finding the equation of a normal

Finding the gradient of the curve at a point, turning it into the normal gradient with the negative reciprocal and writing the normal's equation.

14 question parts: 1 worked-example retry and 13 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: FPM-CAL-05

1. Question 9 · part whole · 5 marks

2021 JUN · Paper 01

Worked-example retry · Attempt part whole only.

9 Find an equation of the normal to the curve with equation

$$y = (x^3 - 2x)e^{(1-x)}$$

at the point on the curve with coordinates $(1, -1)$

(5)

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1. Question 9 · part whole · 5 marks

2021 JUN · Paper 01

Worked-example retry · Attempt part whole only.

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Question 9 continued

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(Total for Question 9 is 5 marks)

1. Question 9 · part whole · 5 marks · Working and answer

Working and answer

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2. Question 4 · part whole · 10 marks

2019 JUN · Paper 01 (regional R)

Further practice · Attempt part whole only.

4 $f(x) = e^{3x}\sqrt{1+2x}$

(a) Show that $f'(x) = \dfrac{2e^{3x}(2+3x)}{\sqrt{1+2x}}$

(4)

(b) Find an equation of the normal to the curve with equation $y = f(x)$ at the point on the curve where $x = 0$

Give your answer in the form $ax + by + c = 0$ where a, b and c are integers.

(6)

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2. Question 4 · part whole · 10 marks

2019 JUN · Paper 01 (regional R)

Further practice · Attempt part whole only.

Question 4 continued

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(Total for Question 4 is 10 marks)

2. Question 4 · part whole · 10 marks · Working and answer

Working and answer

OLevels.com SCHOOL

3. Question 8 · part c · 3 marks

2020 NOV · Paper 01

Further practice · Attempt part c only.

8 Given that $2xy + 5y = e^x$

(a) show that $\frac{dy}{dx} = \frac{y(2x+3)}{2x+5}$

(5)

(b) find the value of $\frac{dy}{dx}$ when $x = 0$

(2)

(c) find an equation of the normal to the curve with equation $2xy + 5y = e^x$ at the point where $x = 0$
Give your answer in the form $px + qy + r = 0$ where p, q and r are integers.

(3)

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3. Question 8 · part c · 3 marks

2020 NOV · Paper 01

Further practice · Attempt part c only.

Question 8 continued

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3. Question 8 · part c · 3 marks

2020 NOV · Paper 01

Further practice · Attempt part c only.

Question 8 continued

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3. Question 8 · part c · 3 marks

2020 NOV · Paper 01

Further practice · Attempt part c only.

Question 8 continued

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(Total for Question 8 is 10 marks)

3. Question 8 · part c · 3 marks · Working and answer

Working and answer

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4. Question 5 · part b · 5 marks

2023 NOV · Paper 02

Further practice · Attempt part b only.

5

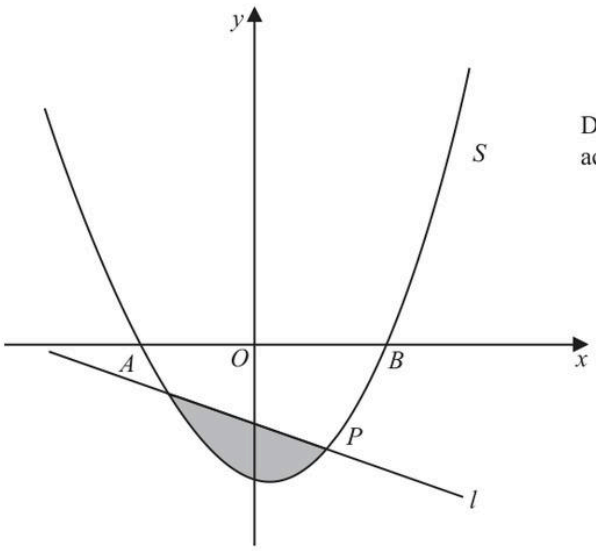


Figure 1

Figure 1 shows part of the curve S with equation $y = px^2 + qx + r$ where p, q and r are constants.

The points A, B and P with coordinates $(-2, 0)$, $(6, 0)$ and $(4, -6)$ respectively lie on S

(a) Show that an equation of S is $y = \frac{x^2}{2} - 2x - 6$

(3)

The line l is the normal to S at the point P

(b) Show that an equation of l is $2y + x + 8 = 0$

(5)

The finite region shown shaded in Figure 1 is bounded by S and l

(c) Use algebraic integration to find the exact area of the shaded region.

(7)

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P 7 3 5 8 6 A 0 1 2 3 2

4. Question 5 · part b · 5 marks · Working and answer

Working and answer

OLevels.com SCHOOL

5. Question 6 · part c · 3 marks

2020 NOV · Paper 02 (regional R)

Further practice · Attempt part c only.

6

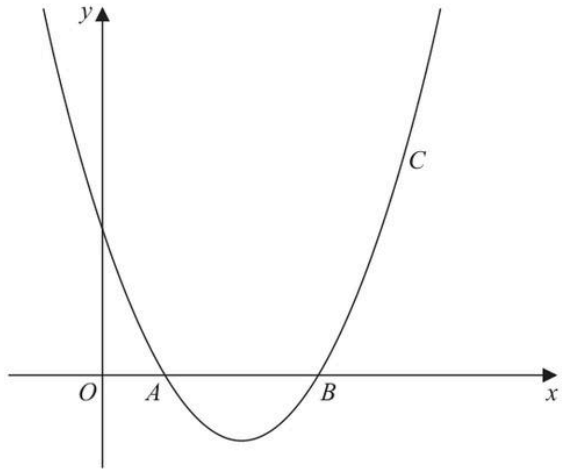


Diagram **NOT** accurately drawn

Figure 3

The curve C with equation $y = x^2 - 5x + 4$ crosses the x-axis at the points A and B, as shown in Figure 3

(a) Find the coordinates of A and the coordinates of B.

(3)

The tangent to C at A meets the tangent to C at B at the point T.

(b) Find the coordinates of T.

(6)

The normal to C at A meets the normal to C at B at the point N.

(c) Find the coordinates of N.

(3)

(d) Find the area of the quadrilateral $ATBN$.

(3)

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5. Question 6 · part c · 3 marks · Working and answer

Working and answer

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6. Question 6 · part b · 4 marks

2022 JAN · Paper 02 (regional R)

Further practice · Attempt part b only.

6 The curve C has equation

$$y = 4\sqrt{x}$$

The point A on C has coordinates (9, 12)

The tangent to C at the point A meets the x-axis at the point T

(a) Find the coordinates of T

(5)

The normal to C at the point A meets the x-axis at the point N

(b) Find the coordinates of N

(4)

(c) Calculate the area of triangle ATN

(2)

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6. Question 6 · part b · 4 marks · Working and answer

Working and answer

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7. Question 11 · part b · 4 marks

2023 JAN · Paper 02

Further practice · Attempt part b only.

11

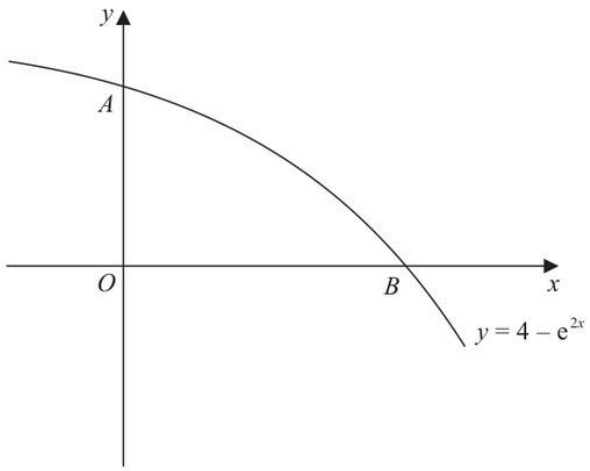


Diagram **NOT** accurately drawn

Figure 3

Figure 3 shows part of the curve C with equation $y = 4 - e^{2x}$
The curve C crosses the y-axis at the point A and the x-axis at the point B.

(a) (i) Write down the y coordinate of point A.

(ii) Show that the x coordinate of B is $x = \ln 2$

(3)

The line l is the normal to C at the point B.

(b) Find an equation for l, giving your answer in the form $y = mx + c$

(4)

The finite region R is bounded by C, l and the y-axis.

(c) Using calculus, find the area of R.
Give your answer to one decimal place.

(7)

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7. Question 11 · part b · 4 marks · Working and answer

Working and answer

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8. Question 7 · part a · 1 mark

2024 JUN · Paper 01

Further practice · Attempt part a only.

7

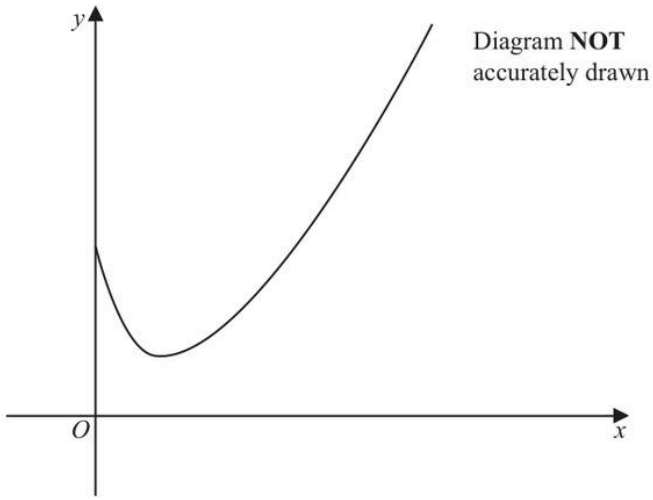


Diagram **NOT** accurately drawn

Figure 1

Figure 1 shows a sketch of part of the curve C with equation

$$y = \frac{x^2}{4} - 3\sqrt{x} + 8$$

The point P lies on C and has coordinates $(4, a)$

(a) Show that $a = 6$

(1)

The line L is the normal to C at the point P

(b) Show that an equation of L is $5y + 4x - 46 = 0$

(6)

The finite region R is bounded by the curve C, the line L, the x-axis and the line with equation $x = 1$

(c) Use calculus to find the exact area of R

(6)

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8. Question 7 · part a · 1 mark · Working and answer

Working and answer

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9. Question 7 · part b · 6 marks

2024 JUN · Paper 01

Further practice · Attempt part b only.

7

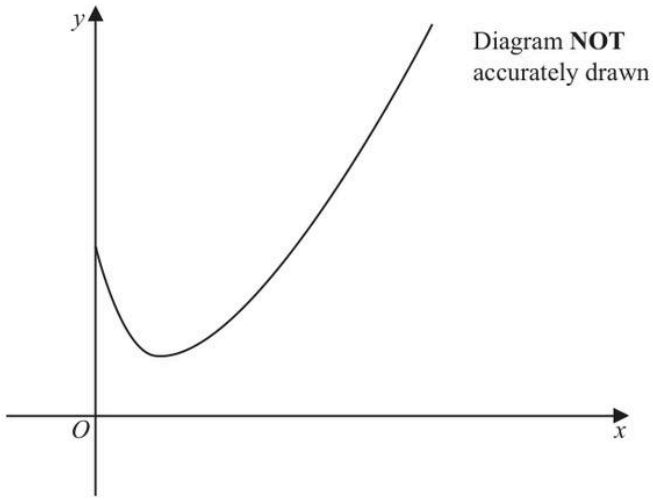


Diagram **NOT** accurately drawn

Figure 1

Figure 1 shows a sketch of part of the curve C with equation

$$y = \frac{x^2}{4} - 3\sqrt{x} + 8$$

The point P lies on C and has coordinates $(4, a)$

(a) Show that $a = 6$

(1)

The line L is the normal to C at the point P

(b) Show that an equation of L is $5y + 4x - 46 = 0$

(6)

The finite region R is bounded by the curve C, the line L, the x-axis and the line with equation $x = 1$

(c) Use calculus to find the exact area of R

(6)

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9. Question 7 · part b · 6 marks · Working and answer

Working and answer

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10. Question 7 · part b · 7 marks

2024 JUN · Paper 02 (regional R)

Further practice · Attempt part b only.

7

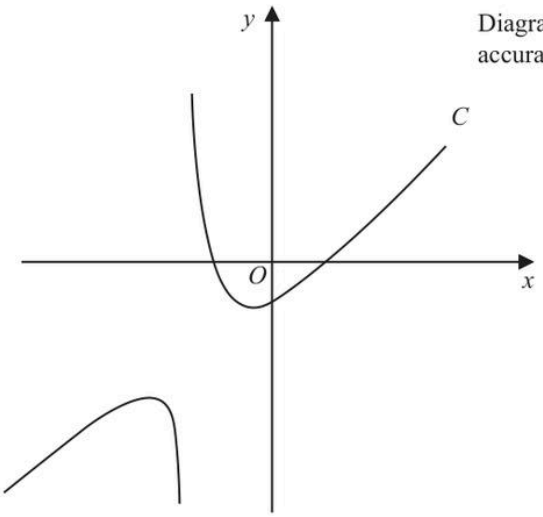


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Figure 2

Figure 2 shows a sketch of part of the curve C with equation

$$y = \frac{x^2 - 1}{4x + 5} \quad \text{where} \quad x \neq -\frac{5}{4}$$

(a) Write down the equation of the asymptote to C that is parallel to the y-axis.

(1)

The line l is the normal to C at the point where $x = -1$

(b) Find an equation of l

(7)

The line l meets C again at the point D

(c) Find the coordinates of D

(6)

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14

P 7 4 2 8 5 R A 0 1 4 3 2

10. Question 7 · part b · 7 marks · Working and answer

Working and answer

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11. Question 4 · part c · 6 marks

2021 NOV · Paper 02

Further practice · Attempt part c only.

4 The curve C has equation $y = 2x^2 + px + q$ where p and q are integers.

The curve C has a stationary point at $(3, -5)$

(a) Show that $p = -12$ and find the value of q

(4)

(b) State, giving a reason, the nature of the stationary point.

(1)

(c) Find an equation of the normal to C at the point on C where $x = 1$

Give your answer in the form $ax + by + c = 0$

(6)

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11. Question 4 · part c · 6 marks

2021 NOV · Paper 02

Further practice · Attempt part c only.

Question 4 continued

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(Total for Question 4 is 11 marks)

11. Question 4 · part c · 6 marks · Working and answer

Working and answer

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12. Question 10 · part d · 6 marks

2023 JUN · Paper 02

Further practice · Attempt part d only.

10 The curve C with equation $y = \dfrac{6 - 3x}{x - 4}$ where $x \neq 4$, crosses the x-axis at the point P and the y-axis at the point Q

(a) Find the coordinates of

(i) P (ii) Q

(2)

(b) Write down an equation of the asymptote to C which is

(i) parallel to the y-axis (ii) parallel to the x-axis

(2)

(c) Sketch C showing clearly the asymptotes and the coordinates of the points P and Q

(3)

The line L is the normal to C at the point on C where $x = 2$

(d) Find an equation of L

(6)

The line L intersects C again at the point R

(e) Find the x coordinate of R

(3)

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12. Question 10 · part d · 6 marks · Working and answer

Working and answer

OLevels.com SCHOOL

13. Question 8 · part whole · 8 marks

2022 JAN · Paper 01 (regional R)

Further practice · Attempt part whole only.

8 The curve C has equation $y = 2x^2 - \sin x$

The point A on C has x coordinate π

Show that an equation of the normal to C at the point A is

$$x + (4\pi + 1)y - \pi(8\pi^2 + 2\pi + 1) = 0$$

(8)

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13. Question 8 · part whole · 8 marks

2022 JAN · Paper 01 (regional R)

Further practice · Attempt part whole only.

Question 8 continued

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(Total for Question 8 is 8 marks)

13. Question 8 · part whole · 8 marks · Working and answer

Working and answer

OLevels.com SCHOOL

14. Question 9 · part b · 7 marks

2025 JUN · Paper 01 (regional R)

Further practice · Attempt part b only.

9 Given that $f(x) = (1+x^2)^4$

(a) show that $f'(x) = 8x(1+x^2)^3$ **(2)**

The curve C has equation $y = \dfrac{\sin 3x}{(1+x^2)^4}$

The point A on C has x coordinate $\dfrac{2\pi}{3}$

(b) Show that the gradient of the normal to C at A is

$$-\frac{1}{3}\left(1+\left(\frac{2\pi}{3}\right)^2\right)^4$$

(7)

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14. Question 9 · part b · 7 marks

2025 JUN · Paper 01 (regional R)

Further practice · Attempt part b only.

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Question 9 continued

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14. Question 9 · part b · 7 marks

2025 JUN · Paper 01 (regional R)

Further practice · Attempt part b only.

Question 9 continued

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14. Question 9 · part b · 7 marks

2025 JUN · Paper 01 (regional R)

Further practice · Attempt part b only.

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Question 9 continued

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(Total for Question 9 is 9 marks)

14. Question 9 · part b · 7 marks · Working and answer

Working and answer

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Practice workbook · Optimisation part 1: building the formula to maximise or minimise

Using a fixed area, volume or perimeter to remove one variable and show a given formula for the quantity to be optimised.

16 question parts: 1 worked-example retry and 15 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: FPM-CAL-10

1. Question 8 · part a · 3 marks

2022 JAN · Paper 01

Worked-example retry · Attempt part a only.

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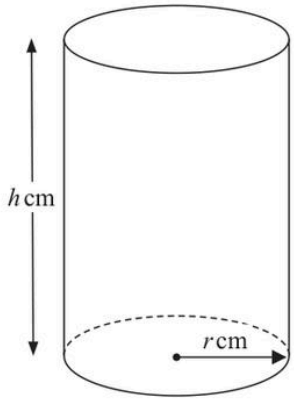


Diagram **NOT** accurately drawn

Figure 1

A solid right circular cylinder has base radius r cm and height h cm as shown in Figure 1.

The cylinder has a volume of $90\pi\,\text{cm}^3$ and a total surface area of $S\,\text{cm}^2$

(a) Show that $S = 2\pi r^2 + \frac{180\pi}{r}$ **(3)**

Given that r can vary,

(b) use calculus to find, to 3 significant figures, the value of r for which S is a minimum, justifying that this value of r gives a minimum value of S **(5)**

(c) Find, to 3 significant figures, the minimum value of S **(2)**

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1. Question 8 · part a · 3 marks · Working and answer

Working and answer

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2. Question 9 · part a · 4 marks

2019 JUN · Paper 01 (regional R)

Further practice · Attempt part a only.

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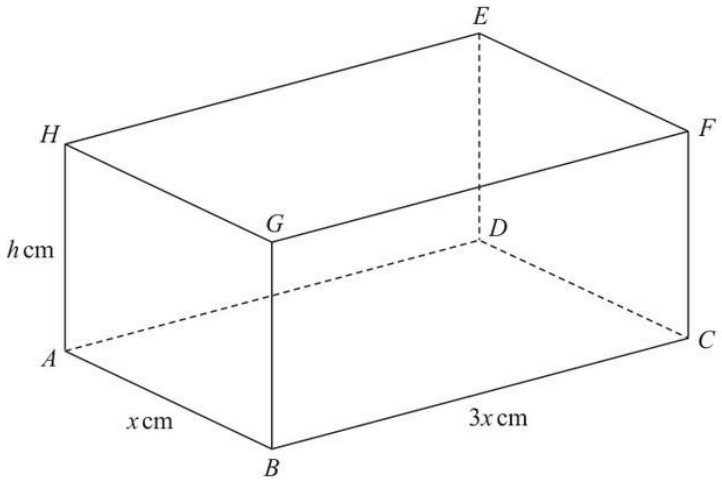


Diagram **NOT** accurately drawn

Figure 3

Figure 3 shows a solid cuboid $ABCDEFGH$

$AB = x$ cm $\quad BC = 3x$ cm $\quad AH = h$ cm

The volume of the cuboid is 540 cm^3

The total surface area of the cuboid is S cm^2

(a) Show that $S = 6x^2 + \frac{1440}{x}$

(4)

Given that x can vary,

(b) use calculus to find, to 3 significant figures, the value of x for which S is a minimum. Justify that this value of x gives a minimum value of S.

(5)

(c) Find, to 3 significant figures, the minimum value of S.

(1)

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24

P 6 1 8 8 2 A 0 2 4 3 6

2. Question 9 · part a · 4 marks · Working and answer

Working and answer

OLevels.com SCHOOL

3. Question 8 · part a · 5 marks

2023 JAN · Paper 01

Further practice · Attempt part a only.

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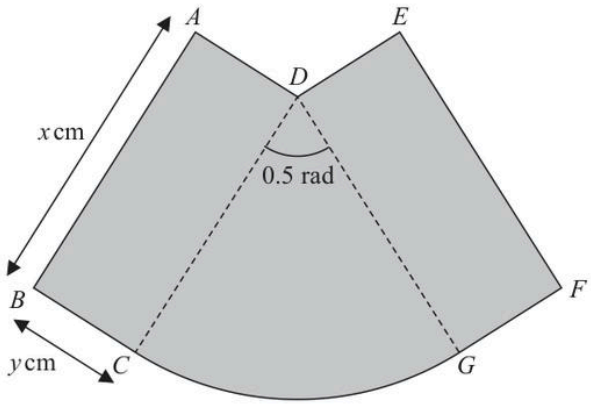


Diagram **NOT** accurately drawn

Figure 1

Figure 1 shows a badge, shown shaded, made from two identical rectangles, $ABCD$ and $DEFG$, and a sector DCG of a circle with centre D.

Each rectangle measures x cm by y cm.
The radius of the sector is x cm and the angle CDG is 0.5 radians.

The area of the badge is 50 cm^2
The perimeter of the badge is P cm.

(a) Show that

$$P = 2x + \frac{100}{x}$$

(5)

Given that x can vary,

(b) use calculus, to find the exact value of x for which P is a minimum.
Justify that this value of x gives a minimum value for P

(6)

(c) Find the minimum value of P
Give your answer in the form $k\sqrt{2}$, where k is an integer to be found.

(2)

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P 7 1 8 1 7 A 0 2 0 3 2

3. Question 8 · part a · 5 marks · Working and answer

Working and answer

OLevels.com SCHOOL

4. Question 7 · part a · 2 marks

2023 NOV · Paper 02

Further practice · Attempt part a only.

7 Two numbers x and y are such that $3x - y = 4$

$S = 5x^3 + y^2$

(a) Show that $S = 5x^3 + 9x^2 - 24x + 16$

(2)

Given that x can vary,

(b) use calculus to find the value of x for which S is a minimum, justifying that this value of x gives a minimum value of S

(5)

(c) Find the minimum value of S

(2)

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4. Question 7 · part a · 2 marks · Working and answer

Working and answer

OLevels.com SCHOOL

5. Question 11 · part a · 3 marks

2023 JAN · Paper 02 (regional R)

Further practice · Attempt part a only.

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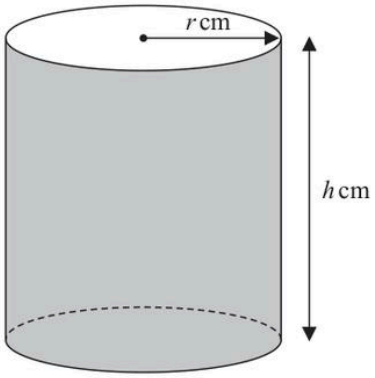


Diagram **NOT** accurately drawn

Figure 4

Figure 4 shows an open container in the shape of a cylinder with radius r cm and height h cm.

Given that the total surface area of the container is $625\pi\,\text{cm}^2$

(a) show that

$$h = \frac{625 - r^2}{2r}$$

(3)

The volume of the container is $V\,\text{cm}^3$

Given that r can vary,

(b) use calculus to find the value, to 3 significant figures, of r for which V is a maximum.
Justify that this value of r gives a maximum value of V

(6)

(c) For the value of r found in part (b), find the corresponding value, to 3 significant figures, of h

(1)

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5. Question 11 · part a · 3 marks · Working and answer

Working and answer

OLevels.com SCHOOL

6. Question 7 · part a · 4 marks

2025 JUN · Paper 02 (regional R)

Further practice · Attempt part a only.

7

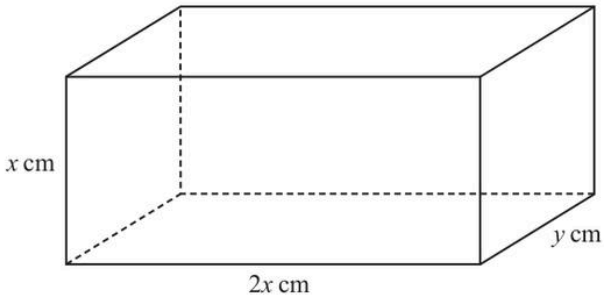


Diagram **NOT** accurately drawn

Figure 3

Figure 3 shows a solid cuboid with sides of length x cm, $2x$ cm and y cm

The volume of the cuboid is 72 cm^3

The total surface area of the cuboid is A cm^2

(a) Show that $A = \frac{216}{x} + 4x^2$ **(4)**

(b) Use calculus to find the minimum value of A
Justify that your value is a minimum value of A **(7)**

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6. Question 7 · part a · 4 marks · Working and answer

Working and answer

OLevels.com SCHOOL

7. Question 6 · part a · 4 marks

2025 JUN · Paper 02

Further practice · Attempt part a only.

6

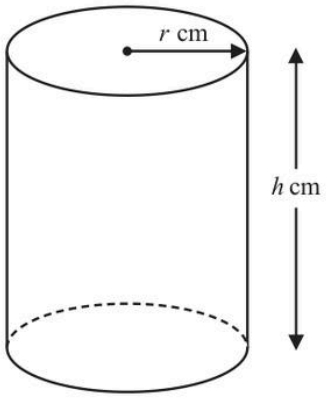


Diagram **NOT** accurately drawn

Figure 1

Figure 1 shows a solid right circular cylinder with radius r cm and height h cm

The total surface area of the cylinder is 700π cm^2

The volume of the cylinder is V cm^3

(a) Show that $V = \pi r\left(350 - r^2\right)$

(4)

Given that r can vary and using calculus,

(b) find, in cm to 3 significant figures, the value of r for which V is a maximum. Justify that this value of r gives a maximum value of V

(5)

(c) Find, to 3 significant figures, the height h cm for which V is a maximum.

(1)

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P 7 2 8 6 7 A 0 1 2 3 2

7. Question 6 · part a · 4 marks · Working and answer

Working and answer

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8. Question 11 · part a · 6 marks

2020 JAN · Paper 01

Further practice · Attempt part a only.

11

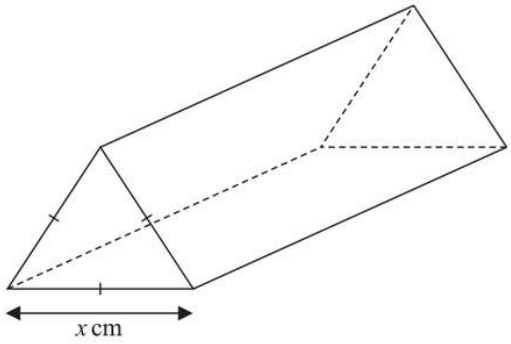


Diagram **NOT** accurately drawn

Figure 4

A company manufactures chocolate bars that are inside packaging that is in the shape of a right triangular prism.

The cross section of the prism is an equilateral triangle with sides of length x cm, as shown in Figure 4.

The volume of the prism is $72\,\text{cm}^3$

The total surface area of the prism is $S\,\text{cm}^2$

(a) Show that

$$S = \frac{\sqrt{3}x^2}{2} + \frac{288\sqrt{3}}{x}$$

(6)

Given that x can vary,

(b) use calculus to find, to 4 significant figures, the value of x for which S is a minimum, justifying that this value gives a minimum value of S.

(5)

(c) Find, to 3 significant figures, the minimum value of S.

(2)

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P 5 9 9 3 8 A 0 3 4 3 6

8. Question 11 · part a · 6 marks · Working and answer

Working and answer

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9. Question 5 · part a · 4 marks

2023 JUN · Paper 01

Further practice · Attempt part a only.

5 A solid cuboid has width x cm, length $4x$ cm and height h cm.
The volume of the cuboid is 75 cm^3 and the surface area of the cuboid is S cm^2

(a) Show that $S = 8x^2 + \frac{375}{2x}$

(4)

Given that x can vary, using calculus,

(b) (i) find to 3 significant figures, the value of x for which S is a minimum,

(ii) justify that this value of x gives a minimum value of S

(5)

(c) Find, to 3 significant figures, the minimum value of S

(2)

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9. Question 5 · part a · 4 marks · Working and answer

Working and answer

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10. Question 8 · part a · 4 marks

2024 JUN · Paper 02

Further practice · Attempt part a only.

8

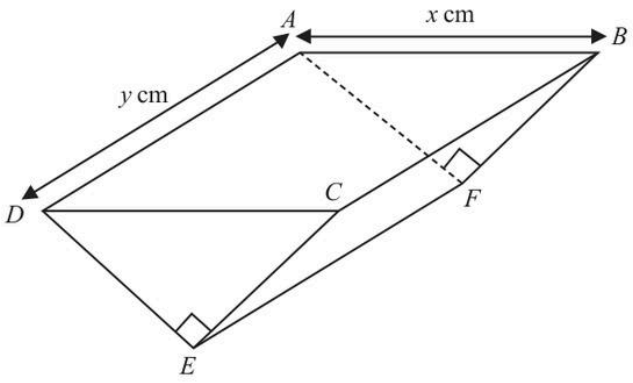


Diagram **NOT** accurately drawn

Figure 4

Figure 4 shows a solid right triangular prism $ABCDEF$

The cross section of the prism is an isosceles triangle.

- $\angle DEC = \angle AFB = 90°$
- $AB = DC = x$ cm
- $AD = BC = FE = y$ cm
- $AF = BF = DE = CE$

The triangular faces of the prism are vertical and the edges AD, BC and FE are horizontal.
The volume of the prism is 3.6 cm^3
The total external surface area of the prism is S cm^2

(a) Show that S satisfies the equation

$$S = \frac{x^2}{2} + \frac{72\left(\sqrt{2}+1\right)}{5x}$$

(4)

Given that x can vary,

(b) use calculus, to find to 3 significant figures, the value of x for which S is a minimum.
Justify that this value of x gives a minimum value of S

(4)

(c) Hence find, to 2 significant figures, the minimum value of S

(2)

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P 7 6 5 0 8 A 0 1 8 3 2

10. Question 8 · part a · 4 marks · Working and answer

Working and answer

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11. Question 5 · part a · 3 marks

2025 NOV · Paper 02

Further practice · Attempt part a only.

5

Diagram **NOT** accurately drawn

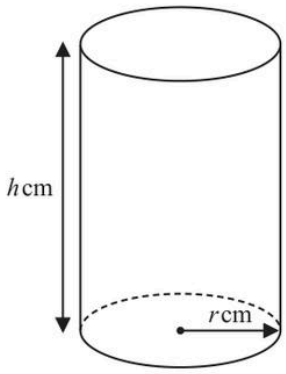


Figure 1

A solid right circular cylinder has base radius r cm and height h cm, as shown in Figure 1

The cylinder has a total surface area of 322π cm^2 and a volume of V cm^3

(a) Show that $V = \pi r\left(161 - r^2\right)$

(3)

Given that r can vary,

(b) use calculus to find, to 3 significant figures, the value of r for which V is a maximum, justifying that this value of r gives a maximum value of V

(5)

(c) Find, to 3 significant figures, the maximum value of V

(2)

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11. Question 5 · part a · 3 marks · Working and answer

Working and answer

OLevels.com SCHOOL

12. Question 11 · part a · 4 marks

2022 JUN · Paper 02 (regional R)

Further practice · Attempt part a only.

11

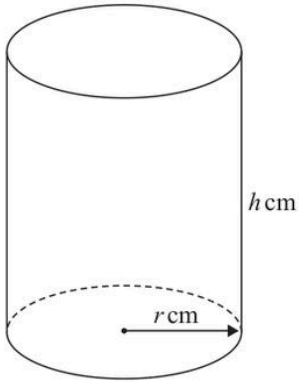


Diagram **NOT** accurately drawn

Figure 3

Figure 3 shows a solid metal right circular cylinder of radius r cm and height h cm.

The total surface area of the cylinder is $600\,\text{cm}^2$

The volume of the cylinder is $V\,\text{cm}^3$

(a) Show that $V = 300r - \pi r^3$ **(4)**

Given that r can vary,

(b) (i) use calculus to show that the exact value of r for which V is a maximum is

$$r = \sqrt{\frac{100}{\pi}}$$

(ii) justify that this value of r gives a maximum value of V **(5)**

The cylinder is melted down and reformed into a sphere of radius p cm.

(c) Find, to one decimal place, the greatest possible value of p **(3)**

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P 7 1 6 4 2 A 0 3 4 3 6

12. Question 11 · part a · 4 marks · Working and answer

Working and answer

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13. Question 3 · part a · 3 marks

2023 JUN · Paper 02 (regional R)

Further practice · Attempt part a only.

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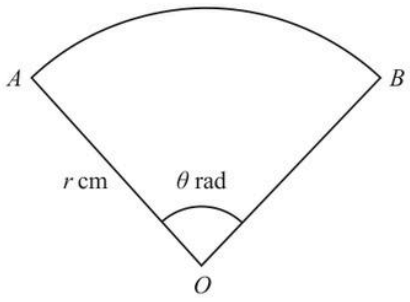


Diagram **NOT** accurately drawn

Figure 1

Figure 1 shows the sector OAB of a circle with centre O.

The radius of the circle is r cm and the angle AOB is θ radians.

The area of the sector is 675 cm^2

(a) Show that the perimeter of the sector, P cm, is given by

$$P = 2r + \frac{1350}{r}$$

(3)

Given that r can vary,

(b) find, using calculus, the minimum value of P
Give your answer in the form $a\sqrt{b}$ where a is an integer and b is a prime number.

(5)

(c) Justify that the value of P you found in (b) is a minimum.

(2)

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P 7 4 2 8 4 A 0 6 3 2

13. Question 3 · part a · 3 marks · Working and answer

Working and answer

OLevels.com SCHOOL

14. Question 7 · part a · 4 marks

2024 NOV · Paper 01

Further practice · Attempt part a only.

7

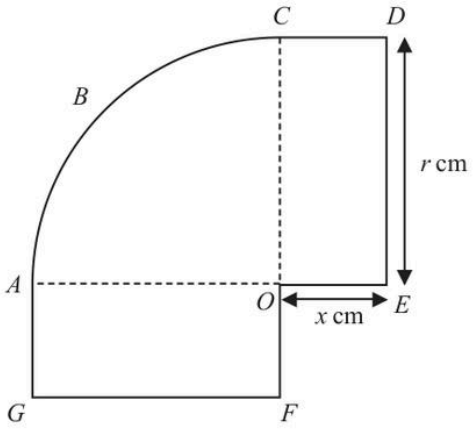


Diagram **NOT** accurately drawn

Figure 2

Figure 2 shows a shape $ABCDEOFG$

$ABCO$ is a quarter circle with radius r cm
$CDEO$ and $AOFG$ are congruent rectangles of length r cm and width x cm

The total area of the shape is $100\,\text{cm}^2$
The perimeter of the shape is P cm

(a) Show that $P = \frac{200}{r} + 2r$

(4)

(b) Use calculus to find the value of r for which P is a minimum, justifying that this value of r gives a minimum value of P

(5)

(c) Find the minimum value of P

(2)

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P 7 3 5 8 5 A 0 1 8 3 2

14. Question 7 · part a · 4 marks · Working and answer

Working and answer

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15. Question 8 · part a · 5 marks

2022 JUN · Paper 02

Further practice · Attempt part a only.

8

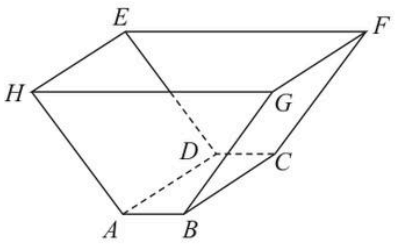


Diagram **NOT** accurately drawn

Figure 2

Figure 2 shows a waste paper basket in the shape of a right prism with 5 faces and a cross section that is a trapezium. The top, $EFGH$, of the waste paper basket is open.

The base of the prism $ABCD$ is a rectangle with

$$AB = DC = 2x\,\text{cm} \quad \text{and} \quad AD = BC = h\,\text{cm}$$

The cross sections $HGBA$ and $EFCD$ are such that

$$EF = HG = 8x\,\text{cm} \quad \text{and} \quad AH = BG = CF = DE = 5x\,\text{cm}$$

The top, $EFGH$, of the waste paper basket is such that

$$EH = FG = h\,\text{cm}$$

The volume of the waste paper basket is $2250\,\text{cm}^3$

The total surface area of the 5 faces of the waste paper basket is $S\,\text{cm}^2$

(a) Show that $S = 40x^2 + \dfrac{1350}{x}$

(5)

Given that x can vary,

(b) use calculus, to find, to 3 significant figures, the value of x for which S is a minimum.
Justify that this value of x gives a minimum value of S

(5)

(c) Find, to 3 significant figures, the minimum value of S

(2)

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15. Question 8 · part a · 5 marks · Working and answer

Working and answer

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16. Question 7 · part a · 5 marks

2022 JAN · Paper 01 (regional R)

Further practice · Attempt part a only.

7

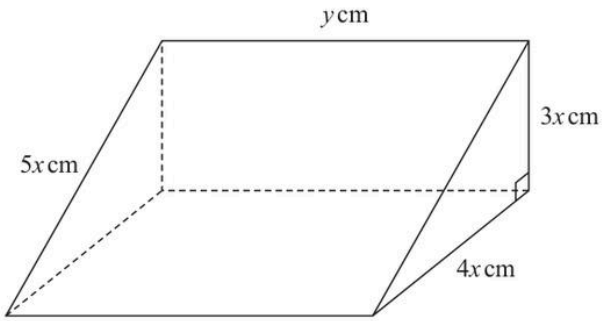


Diagram **NOT** accurately drawn

Figure 2

Figure 2 shows a block of wood in the shape of a right triangular prism.

The cross section of the prism is a right-angled triangle with sides of length $3x$ cm, $4x$ cm and $5x$ cm.

The length of the prism is y cm.

The total surface area of the five faces of the prism is 144 cm^2

The volume of the prism is V cm^3

(a) Show that

$$V = 72x - 6x^3$$

(5)

Given that x can vary,

(b) use calculus to find the value of x for which V is a maximum, justifying that this value gives a maximum value of V

(4)

(c) Find the maximum value of V

(2)

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16. Question 7 · part a · 5 marks · Working and answer

Working and answer

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Practice workbook · Optimisation part 2: finding, justifying and evaluating the maximum or minimum

Differentiating the single-variable expression, solving dV/dx = 0 for the critical value and using the second derivative to justify a maximum or minimum.

33 question parts: 1 worked-example retry and 32 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: FPM-CAL-11

1. Question 8 · part b · 5 marks

2022 JAN · Paper 01

Worked-example retry · Attempt part b only.

8

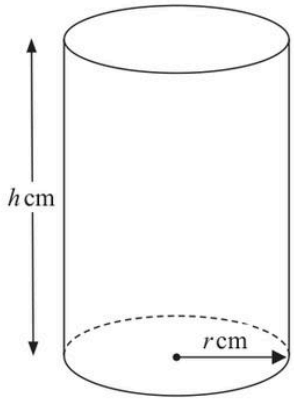


Diagram **NOT** accurately drawn

Figure 1

A solid right circular cylinder has base radius r cm and height h cm as shown in Figure 1.

The cylinder has a volume of $90\pi\,\text{cm}^3$ and a total surface area of $S\,\text{cm}^2$

(a) Show that $S = 2\pi r^2 + \dfrac{180\pi}{r}$ (3)

Given that r can vary,

(b) use calculus to find, to 3 significant figures, the value of r for which S is a minimum, justifying that this value of r gives a minimum value of S (5)

(c) Find, to 3 significant figures, the minimum value of S (2)

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P 6 6 3 1 0 A 0 2 0 3 6

1. Question 8 · part b · 5 marks · Working and answer

Working and answer

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2. Question 9 · part b · 5 marks

2019 JUN · Paper 01 (regional R)

Further practice · Attempt part b only.

9

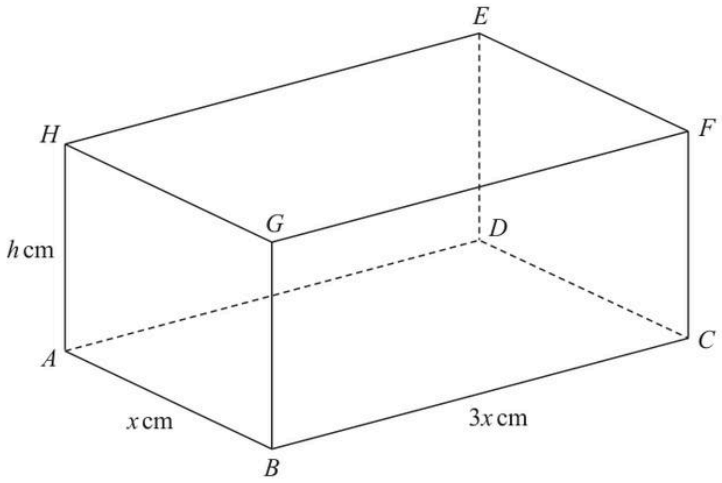


Diagram **NOT** accurately drawn

Figure 3

Figure 3 shows a solid cuboid $ABCDEFGH$

$AB = x$ cm $\quad BC = 3x$ cm $\quad AH = h$ cm

The volume of the cuboid is $540\,\text{cm}^3$

The total surface area of the cuboid is $S\,\text{cm}^2$

(a) Show that $S = 6x^2 + \frac{1440}{x}$

(4)

Given that x can vary,

(b) use calculus to find, to 3 significant figures, the value of x for which S is a minimum. Justify that this value of x gives a minimum value of S.

(5)

(c) Find, to 3 significant figures, the minimum value of S.

(1)

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24

P 6 1 8 8 2 A 0 2 4 3 6

2. Question 9 · part b · 5 marks · Working and answer

Working and answer

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3. Question 8 · part b · 6 marks

2023 JAN · Paper 01

Further practice · Attempt part b only.

8

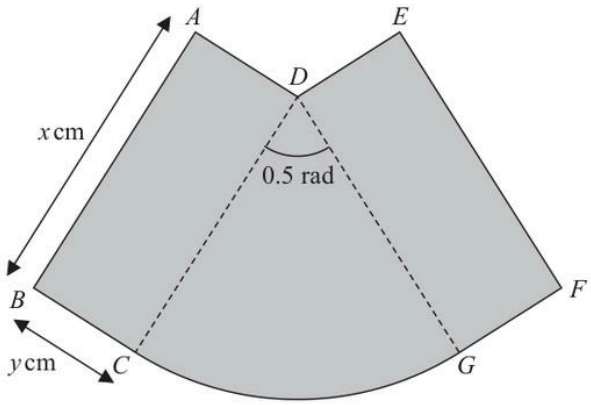


Diagram **NOT** accurately drawn

Figure 1

Figure 1 shows a badge, shown shaded, made from two identical rectangles, $ABCD$ and $DEFG$, and a sector DCG of a circle with centre D.

Each rectangle measures x cm by y cm.
The radius of the sector is x cm and the angle CDG is 0.5 radians.

The area of the badge is 50 cm^2
The perimeter of the badge is P cm.

(a) Show that

$$P = 2x + \frac{100}{x}$$

(5)

Given that x can vary,

(b) use calculus, to find the exact value of x for which P is a minimum.
Justify that this value of x gives a minimum value for P

(6)

(c) Find the minimum value of P
Give your answer in the form $k\sqrt{2}$, where k is an integer to be found.

(2)

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20

P 7 1 8 1 7 A 0 2 0 3 2

3. Question 8 · part b · 6 marks · Working and answer

Working and answer

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4. Question 7 · part b · 5 marks

2023 NOV · Paper 02

Further practice · Attempt part b only.

7 Two numbers x and y are such that $3x - y = 4$

$S = 5x^3 + y^2$

(a) Show that $S = 5x^3 + 9x^2 - 24x + 16$

(2)

Given that x can vary,

(b) use calculus to find the value of x for which S is a minimum, justifying that this value of x gives a minimum value of S

(5)

(c) Find the minimum value of S

(2)

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4. Question 7 · part b · 5 marks · Working and answer

Working and answer

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5. Question 11 · part b · 6 marks

2023 JAN · Paper 02 (regional R)

Further practice · Attempt part b only.

11

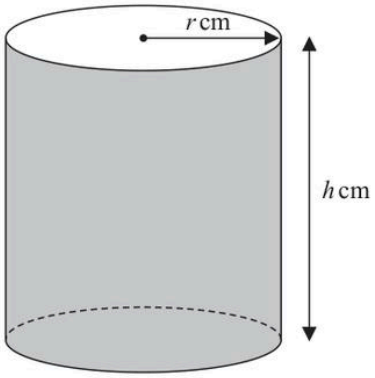


Diagram **NOT** accurately drawn

Figure 4

Figure 4 shows an open container in the shape of a cylinder with radius r cm and height h cm.

Given that the total surface area of the container is $625\pi\,\text{cm}^2$

(a) show that

$$h = \frac{625 - r^2}{2r}$$

(3)

The volume of the container is $V\,\text{cm}^3$

Given that r can vary,

(b) use calculus to find the value, to 3 significant figures, of r for which V is a maximum.
Justify that this value of r gives a maximum value of V

(6)

(c) For the value of r found in part (b), find the corresponding value, to 3 significant figures, of h

(1)

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5. Question 11 · part b · 6 marks · Working and answer

Working and answer

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6. Question 7 · part b · 7 marks

2025 JUN · Paper 02 (regional R)

Further practice · Attempt part b only.

7

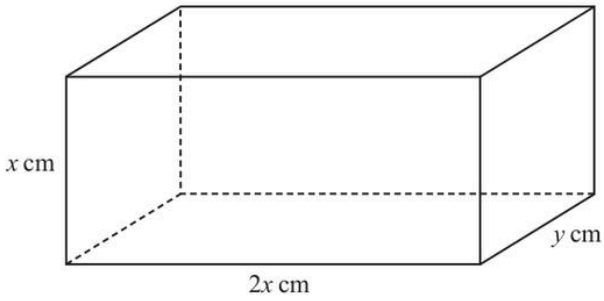


Diagram **NOT** accurately drawn

Figure 3

Figure 3 shows a solid cuboid with sides of length x cm, $2x$ cm and y cm

The volume of the cuboid is 72 cm^3

The total surface area of the cuboid is A cm^2

(a) Show that $A = \frac{216}{x} + 4x^2$

(4)

(b) Use calculus to find the minimum value of A
Justify that your value is a minimum value of A

(7)

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6. Question 7 · part b · 7 marks

2025 JUN · Paper 02 (regional R)

Further practice · Attempt part b only.

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Question 7 continued

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6. Question 7 · part b · 7 marks

2025 JUN · Paper 02 (regional R)

Further practice · Attempt part b only.

Question 7 continued

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6. Question 7 · part b · 7 marks

2025 JUN · Paper 02 (regional R)

Further practice · Attempt part b only.

Question 7 continued

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(Total for Question 7 is 11 marks)

6. Question 7 · part b · 7 marks · Working and answer

Working and answer

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7. Question 6 · part b · 5 marks

2025 JUN · Paper 02

Further practice · Attempt part b only.

6

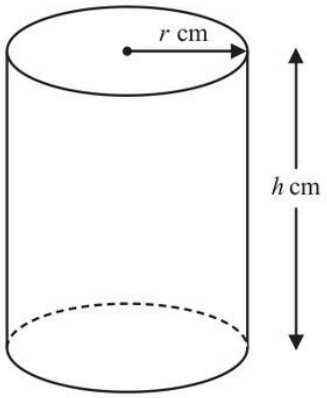


Diagram **NOT** accurately drawn

Figure 1

Figure 1 shows a solid right circular cylinder with radius r cm and height h cm

The total surface area of the cylinder is 700π cm^2

The volume of the cylinder is V cm^3

(a) Show that $V = \pi r\left(350 - r^2\right)$ **(4)**

Given that r can vary and using calculus,

(b) find, in cm to 3 significant figures, the value of r for which V is a maximum. Justify that this value of r gives a maximum value of V **(5)**

(c) Find, to 3 significant figures, the height h cm for which V is a maximum. **(1)**

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P 7 2 8 6 7 A 0 1 2 3 2

7. Question 6 · part b · 5 marks · Working and answer

Working and answer

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8. Question 11 · part b · 5 marks

2020 JAN · Paper 01

Further practice · Attempt part b only.

11

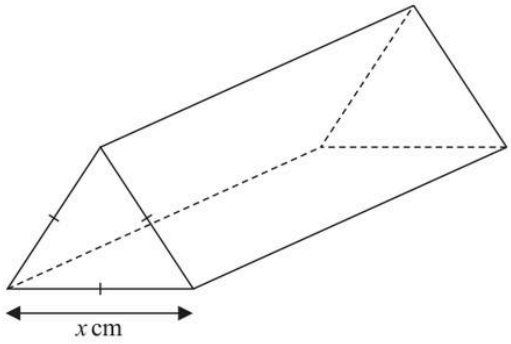


Diagram **NOT** accurately drawn

Figure 4

A company manufactures chocolate bars that are inside packaging that is in the shape of a right triangular prism.

The cross section of the prism is an equilateral triangle with sides of length x cm, as shown in Figure 4.

The volume of the prism is $72\,\text{cm}^3$

The total surface area of the prism is $S\,\text{cm}^2$

(a) Show that

$$S = \frac{\sqrt{3}x^2}{2} + \frac{288\sqrt{3}}{x}$$

(6)

Given that x can vary,

(b) use calculus to find, to 4 significant figures, the value of x for which S is a minimum, justifying that this value gives a minimum value of S.

(5)

(c) Find, to 3 significant figures, the minimum value of S.

(2)

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8. Question 11 · part b · 5 marks · Working and answer

Working and answer

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9. Question 5 · part b.i · 0 marks

2023 JUN · Paper 01

Further practice · Attempt part b.i only.

5 A solid cuboid has width x cm, length $4x$ cm and height h cm.
The volume of the cuboid is 75 cm^3 and the surface area of the cuboid is S cm^2

(a) Show that $S = 8x^2 + \frac{375}{2x}$

(4)

Given that x can vary, using calculus,

(b) (i) find to 3 significant figures, the value of x for which S is a minimum,

(ii) justify that this value of x gives a minimum value of S

(5)

(c) Find, to 3 significant figures, the minimum value of S

(2)

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9. Question 5 · part b.i · 0 marks · Working and answer

Working and answer

OLevels.com SCHOOL

10. Question 5 · part b.ii · 5 marks

2023 JUN · Paper 01

Further practice · Attempt part b.ii only.

5 A solid cuboid has width x cm, length $4x$ cm and height h cm.
The volume of the cuboid is $75\,\text{cm}^3$ and the surface area of the cuboid is $S\,\text{cm}^2$

(a) Show that $S = 8x^2 + \frac{375}{2x}$

(4)

Given that x can vary, using calculus,

(b) (i) find to 3 significant figures, the value of x for which S is a minimum,

(ii) justify that this value of x gives a minimum value of S

(5)

(c) Find, to 3 significant figures, the minimum value of S

(2)

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10. Question 5 · part b.ii · 5 marks · Working and answer

Working and answer

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11. Question 8 · part b · 4 marks

2024 JUN · Paper 02

Further practice · Attempt part b only.

8

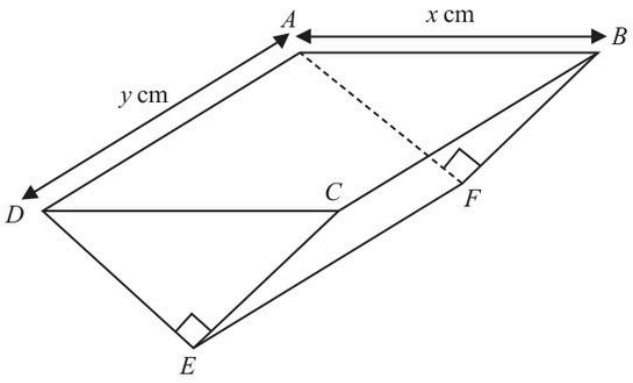


Diagram **NOT** accurately drawn

Figure 4

Figure 4 shows a solid right triangular prism $ABCDEF$

The cross section of the prism is an isosceles triangle.

- $\angle DEC = \angle AFB = 90°$
- $AB = DC = x$ cm
- $AD = BC = FE = y$ cm
- $AF = BF = DE = CE$

The triangular faces of the prism are vertical and the edges AD, BC and FE are horizontal.
The volume of the prism is 3.6 cm^3
The total external surface area of the prism is S cm^2

(a) Show that S satisfies the equation

$$S = \frac{x^2}{2} + \frac{72\left(\sqrt{2}+1\right)}{5x}$$

(4)

Given that x can vary,

(b) use calculus, to find to 3 significant figures, the value of x for which S is a minimum.
Justify that this value of x gives a minimum value of S

(4)

(c) Hence find, to 2 significant figures, the minimum value of S

(2)

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11. Question 8 · part b · 4 marks · Working and answer

Working and answer

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12. Question 5 · part b · 5 marks

2025 NOV · Paper 02

Further practice · Attempt part b only.

5

Diagram **NOT** accurately drawn

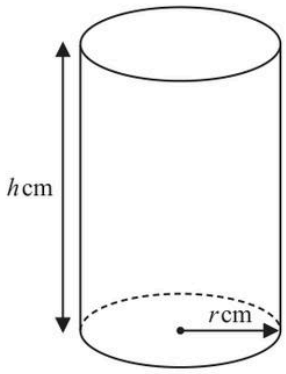


Figure 1

A solid right circular cylinder has base radius r cm and height h cm, as shown in Figure 1

The cylinder has a total surface area of 322π cm^2 and a volume of V cm^3

(a) Show that $V = \pi r\left(161 - r^2\right)$

(3)

Given that r can vary,

(b) use calculus to find, to 3 significant figures, the value of r for which V is a maximum, justifying that this value of r gives a maximum value of V

(5)

(c) Find, to 3 significant figures, the maximum value of V

(2)

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12. Question 5 · part b · 5 marks · Working and answer

Working and answer

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13. Question 11 · part b.i · 0 marks

2022 JUN · Paper 02 (regional R)

Further practice · Attempt part b.i only.

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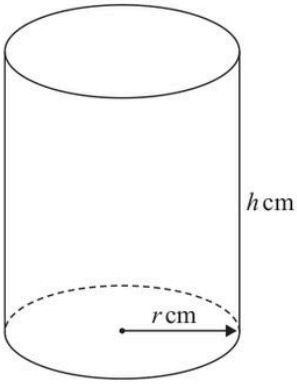


Diagram **NOT** accurately drawn

Figure 3

Figure 3 shows a solid metal right circular cylinder of radius r cm and height h cm.

The total surface area of the cylinder is $600\,\text{cm}^2$

The volume of the cylinder is $V\,\text{cm}^3$

(a) Show that $V = 300r - \pi r^3$ **(4)**

Given that r can vary,

(b) (i) use calculus to show that the exact value of r for which V is a maximum is

$$r = \sqrt{\frac{100}{\pi}}$$

(ii) justify that this value of r gives a maximum value of V **(5)**

The cylinder is melted down and reformed into a sphere of radius p cm.

(c) Find, to one decimal place, the greatest possible value of p **(3)**

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13. Question 11 · part b.i · 0 marks · Working and answer

Working and answer

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14. Question 11 · part b.ii · 5 marks

2022 JUN · Paper 02 (regional R)

Further practice · Attempt part b.ii only.

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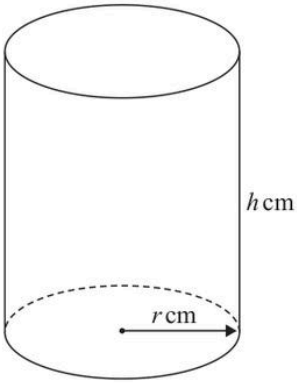


Diagram **NOT** accurately drawn

Figure 3

Figure 3 shows a solid metal right circular cylinder of radius r cm and height h cm.

The total surface area of the cylinder is $600\,\text{cm}^2$

The volume of the cylinder is $V\,\text{cm}^3$

(a) Show that $V = 300r - \pi r^3$ **(4)**

Given that r can vary,

(b) (i) use calculus to show that the exact value of r for which V is a maximum is

$$r = \sqrt{\frac{100}{\pi}}$$

(ii) justify that this value of r gives a maximum value of V **(5)**

The cylinder is melted down and reformed into a sphere of radius p cm.

(c) Find, to one decimal place, the greatest possible value of p **(3)**

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14. Question 11 · part b.ii · 5 marks · Working and answer

Working and answer

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15. Question 3 · part b · 5 marks

2023 JUN · Paper 02 (regional R)

Further practice · Attempt part b only.

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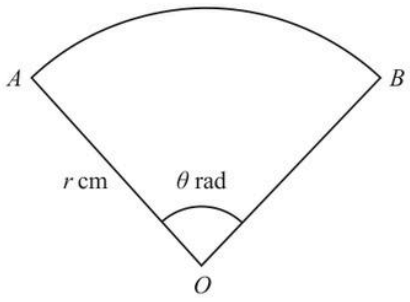


Diagram **NOT** accurately drawn

Figure 1

Figure 1 shows the sector OAB of a circle with centre O.

The radius of the circle is r cm and the angle AOB is θ radians.

The area of the sector is 675 cm^2

(a) Show that the perimeter of the sector, P cm, is given by

$$P = 2r + \frac{1350}{r}$$

(3)

Given that r can vary,

(b) find, using calculus, the minimum value of P
Give your answer in the form $a\sqrt{b}$ where a is an integer and b is a prime number.

(5)

(c) Justify that the value of P you found in (b) is a minimum.

(2)

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15. Question 3 · part b · 5 marks · Working and answer

Working and answer

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16. Question 3 · part c · 2 marks

2023 JUN · Paper 02 (regional R)

Further practice · Attempt part c only.

3

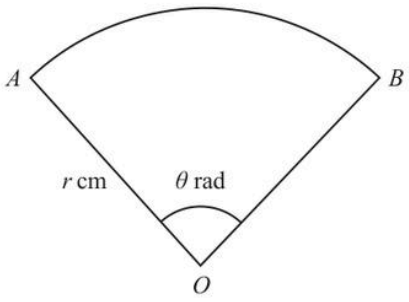


Diagram **NOT** accurately drawn

Figure 1

Figure 1 shows the sector OAB of a circle with centre O.

The radius of the circle is r cm and the angle AOB is θ radians.

The area of the sector is 675 cm^2

(a) Show that the perimeter of the sector, P cm, is given by

$$P = 2r + \frac{1350}{r}$$

(3)

Given that r can vary,

(b) find, using calculus, the minimum value of P
Give your answer in the form $a\sqrt{b}$ where a is an integer and b is a prime number.

(5)

(c) Justify that the value of P you found in (b) is a minimum.

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16. Question 3 · part c · 2 marks

2023 JUN · Paper 02 (regional R)

Further practice · Attempt part c only.

Question 3 continued

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(Total for Question 3 is 10 marks)

16. Question 3 · part c · 2 marks · Working and answer

Working and answer

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17. Question 7 · part b · 5 marks

2024 NOV · Paper 01

Further practice · Attempt part b only.

7

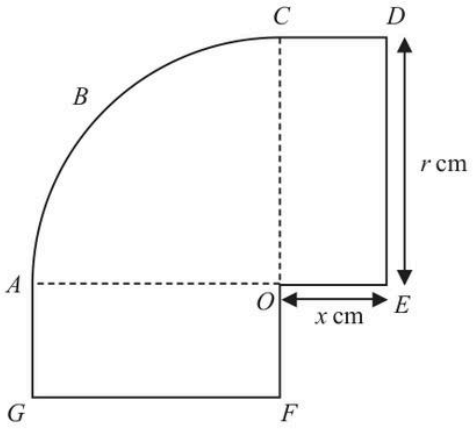


Diagram **NOT** accurately drawn

Figure 2

Figure 2 shows a shape $ABCDEOFG$

$ABCO$ is a quarter circle with radius r cm
$CDEO$ and $AOFG$ are congruent rectangles of length r cm and width x cm

The total area of the shape is $100\,\text{cm}^2$
The perimeter of the shape is P cm

(a) Show that $P = \frac{200}{r} + 2r$

(4)

(b) Use calculus to find the value of r for which P is a minimum, justifying that this value of r gives a minimum value of P

(5)

(c) Find the minimum value of P

(2)

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17. Question 7 · part b · 5 marks · Working and answer

Working and answer

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18. Question 8 · part b · 5 marks

2022 JUN · Paper 02

Further practice · Attempt part b only.

8

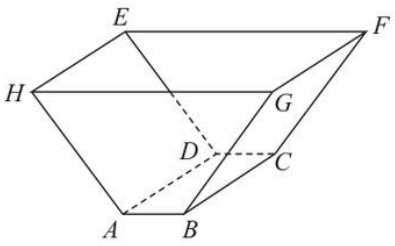


Diagram **NOT** accurately drawn

Figure 2

Figure 2 shows a waste paper basket in the shape of a right prism with 5 faces and a cross section that is a trapezium. The top, $EFGH$, of the waste paper basket is open.

The base of the prism $ABCD$ is a rectangle with

$$AB = DC = 2x\,\text{cm} \text{ and } AD = BC = h\,\text{cm}$$

The cross sections $HGBA$ and $EFCD$ are such that

$$EF = HG = 8x\,\text{cm} \text{ and } AH = BG = CF = DE = 5x\,\text{cm}$$

The top, $EFGH$, of the waste paper basket is such that

$$EH = FG = h\,\text{cm}$$

The volume of the waste paper basket is $2250\,\text{cm}^3$

The total surface area of the 5 faces of the waste paper basket is $S\,\text{cm}^2$

(a) Show that $S = 40x^2 + \dfrac{1350}{x}$

(5)

Given that x can vary,

(b) use calculus, to find, to 3 significant figures, the value of x for which S is a minimum.
Justify that this value of x gives a minimum value of S

(5)

(c) Find, to 3 significant figures, the minimum value of S

(2)

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18. Question 8 · part b · 5 marks · Working and answer

Working and answer

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19. Question 7 · part b · 4 marks

2022 JAN · Paper 01 (regional R)

Further practice · Attempt part b only.

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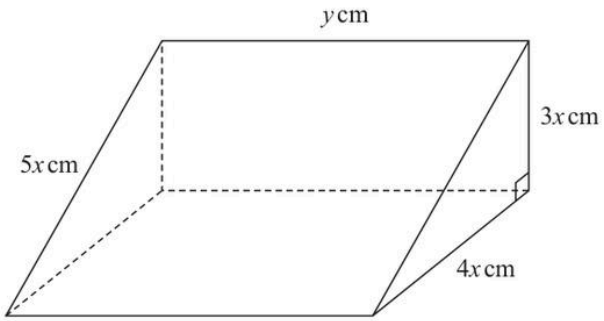


Diagram **NOT** accurately drawn

Figure 2

Figure 2 shows a block of wood in the shape of a right triangular prism.

The cross section of the prism is a right-angled triangle with sides of length $3x$ cm, $4x$ cm and $5x$ cm.

The length of the prism is y cm.

The total surface area of the five faces of the prism is 144 cm^2

The volume of the prism is V cm^3

(a) Show that

$$V = 72x - 6x^3$$

(5)

Given that x can vary,

(b) use calculus to find the value of x for which V is a maximum, justifying that this value gives a maximum value of V

(4)

(c) Find the maximum value of V

(2)

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19. Question 7 · part b · 4 marks · Working and answer

Working and answer

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20. Question 9 · part c · 1 mark

2019 JUN · Paper 01 (regional R)

Further practice · Attempt part c only.

9

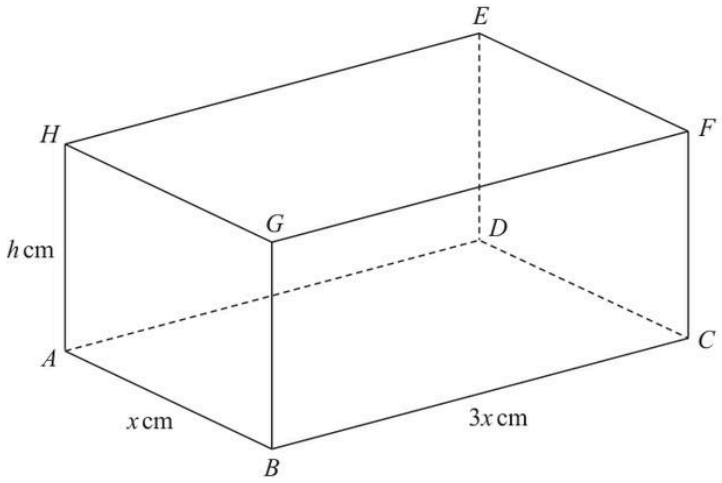


Diagram **NOT** accurately drawn

Figure 3

Figure 3 shows a solid cuboid $ABCDEFGH$

$AB = x$ cm $\quad BC = 3x$ cm $\quad AH = h$ cm

The volume of the cuboid is $540\,\text{cm}^3$

The total surface area of the cuboid is $S\,\text{cm}^2$

(a) Show that $S = 6x^2 + \frac{1440}{x}$

(4)

Given that x can vary,

(b) use calculus to find, to 3 significant figures, the value of x for which S is a minimum. Justify that this value of x gives a minimum value of S.

(5)

(c) Find, to 3 significant figures, the minimum value of S.

(1)

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20. Question 9 · part c · 1 mark

2019 JUN · Paper 01 (regional R)

Further practice · Attempt part c only.

Question 9 continued

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20. Question 9 · part c · 1 mark

2019 JUN · Paper 01 (regional R)

Further practice · Attempt part c only.

Question 9 continued

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20. Question 9 · part c · 1 mark

2019 JUN · Paper 01 (regional R)

Further practice · Attempt part c only.

Question 9 continued

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(Total for Question 9 is 10 marks)

20. Question 9 · part c · 1 mark · Working and answer

Working and answer

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21. Question 8 · part c · 2 marks

2022 JAN · Paper 01

Further practice · Attempt part c only.

8

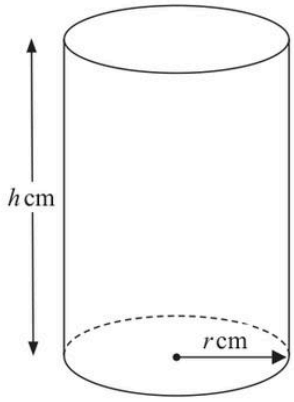


Diagram **NOT** accurately drawn

Figure 1

A solid right circular cylinder has base radius r cm and height h cm as shown in Figure 1.

The cylinder has a volume of $90\pi\,\text{cm}^3$ and a total surface area of $S\,\text{cm}^2$

(a) Show that $S = 2\pi r^2 + \frac{180\pi}{r}$ (3)

Given that r can vary,

(b) use calculus to find, to 3 significant figures, the value of r for which S is a minimum, justifying that this value of r gives a minimum value of S (5)

(c) Find, to 3 significant figures, the minimum value of S (2)

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21. Question 8 · part c · 2 marks

2022 JAN · Paper 01

Further practice · Attempt part c only.

Question 8 continued

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21

Turn over

21. Question 8 · part c · 2 marks

2022 JAN · Paper 01

Further practice · Attempt part c only.

Question 8 continued

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21. Question 8 · part c · 2 marks

2022 JAN · Paper 01

Further practice · Attempt part c only.

Question 8 continued

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(Total for Question 8 is 10 marks)

21. Question 8 · part c · 2 marks · Working and answer

Working and answer

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22. Question 8 · part c · 2 marks

2023 JAN · Paper 01

Further practice · Attempt part c only.

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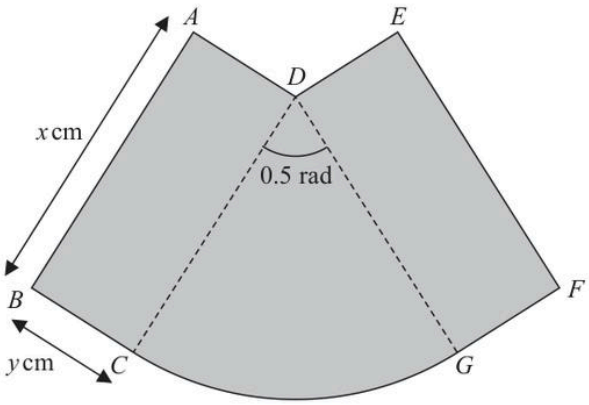


Diagram **NOT** accurately drawn

Figure 1

Figure 1 shows a badge, shown shaded, made from two identical rectangles, $ABCD$ and $DEFG$, and a sector DCG of a circle with centre D.

Each rectangle measures x cm by y cm.
The radius of the sector is x cm and the angle CDG is 0.5 radians.

The area of the badge is 50 cm^2
The perimeter of the badge is P cm.

(a) Show that

$$P = 2x + \frac{100}{x}$$

(5)

Given that x can vary,

(b) use calculus, to find the exact value of x for which P is a minimum.
Justify that this value of x gives a minimum value for P

(6)

(c) Find the minimum value of P
Give your answer in the form $k\sqrt{2}$, where k is an integer to be found.

(2)

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22. Question 8 · part c · 2 marks

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Further practice · Attempt part c only.

Question 8 continued

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22. Question 8 · part c · 2 marks

2023 JAN · Paper 01

Further practice · Attempt part c only.

Question 8 continued

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22. Question 8 · part c · 2 marks

2023 JAN · Paper 01

Further practice · Attempt part c only.

Question 8 continued

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(Total for Question 8 is 13 marks)

22. Question 8 · part c · 2 marks · Working and answer

Working and answer

OLevels.com SCHOOL

23. Question 7 · part c · 2 marks

2023 NOV · Paper 02

Further practice · Attempt part c only.

7 Two numbers x and y are such that $3x - y = 4$

$S = 5x^3 + y^2$

(a) Show that $S = 5x^3 + 9x^2 - 24x + 16$

(2)

Given that x can vary,

(b) use calculus to find the value of x for which S is a minimum, justifying that this value of x gives a minimum value of S

(5)

(c) Find the minimum value of S

(2)

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23. Question 7 · part c · 2 marks

2023 NOV · Paper 02

Further practice · Attempt part c only.

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Question 7 continued

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(Total for Question 7 is 9 marks)

23. Question 7 · part c · 2 marks · Working and answer

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24. Question 11 · part c · 1 mark

2023 JAN · Paper 02 (regional R)

Further practice · Attempt part c only.

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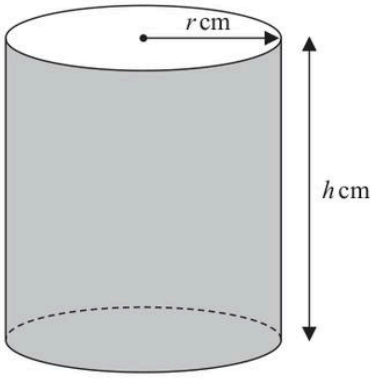


Diagram **NOT** accurately drawn

Figure 4

Figure 4 shows an open container in the shape of a cylinder with radius r cm and height h cm.

Given that the total surface area of the container is $625\pi\,\text{cm}^2$

(a) show that

$$h = \frac{625 - r^2}{2r}$$

(3)

The volume of the container is $V\,\text{cm}^3$

Given that r can vary,

(b) use calculus to find the value, to 3 significant figures, of r for which V is a maximum.
Justify that this value of r gives a maximum value of V

(6)

(c) For the value of r found in part (b), find the corresponding value, to 3 significant figures, of h

(1)

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24. Question 11 · part c · 1 mark

2023 JAN · Paper 02 (regional R)

Further practice · Attempt part c only.

Question 11 continued

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24. Question 11 · part c · 1 mark

2023 JAN · Paper 02 (regional R)

Further practice · Attempt part c only.

Question 11 continued

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24. Question 11 · part c · 1 mark

2023 JAN · Paper 02 (regional R)

Further practice · Attempt part c only.

Question 11 continued

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OLevels.com SCHOOL

24. Question 11 · part c · 1 mark

2023 JAN · Paper 02 (regional R)

Further practice · Attempt part c only.

Question 11 continued

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(Total for Question 11 is 10 marks)

24. Question 11 · part c · 1 mark · Working and answer

Working and answer

OLevels.com SCHOOL

25. Question 6 · part c · 1 mark

2025 JUN · Paper 02

Further practice · Attempt part c only.

6

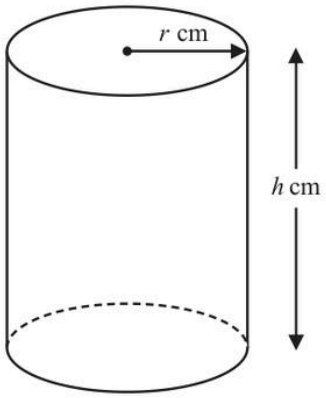


Diagram **NOT** accurately drawn

Figure 1

Figure 1 shows a solid right circular cylinder with radius r cm and height h cm

The total surface area of the cylinder is 700π cm^2

The volume of the cylinder is V cm^3

(a) Show that $V = \pi r\left(350 - r^2\right)$ **(4)**

Given that r can vary and using calculus,

(b) find, in cm to 3 significant figures, the value of r for which V is a maximum. Justify that this value of r gives a maximum value of V **(5)**

(c) Find, to 3 significant figures, the height h cm for which V is a maximum. **(1)**

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25. Question 6 · part c · 1 mark

2025 JUN · Paper 02

Further practice · Attempt part c only.

Question 6 continued

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(Total for Question 6 is 10 marks)

25. Question 6 · part c · 1 mark · Working and answer

Working and answer

OLevels.com SCHOOL

26. Question 11 · part c · 2 marks

2020 JAN · Paper 01

Further practice · Attempt part c only.

11

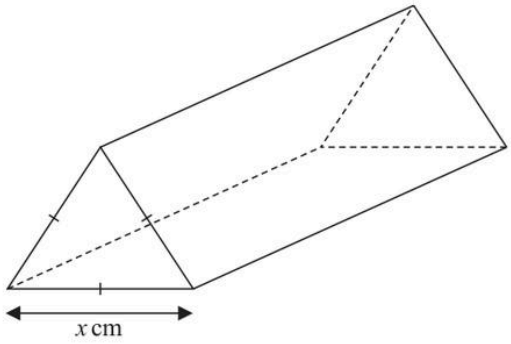


Diagram **NOT** accurately drawn

Figure 4

A company manufactures chocolate bars that are inside packaging that is in the shape of a right triangular prism.

The cross section of the prism is an equilateral triangle with sides of length x cm, as shown in Figure 4.

The volume of the prism is $72\,\text{cm}^3$

The total surface area of the prism is $S\,\text{cm}^2$

(a) Show that

$$S = \frac{\sqrt{3}x^2}{2} + \frac{288\sqrt{3}}{x}$$

(6)

Given that x can vary,

(b) use calculus to find, to 4 significant figures, the value of x for which S is a minimum, justifying that this value gives a minimum value of S.

(5)

(c) Find, to 3 significant figures, the minimum value of S.

(2)

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26. Question 11 · part c · 2 marks

2020 JAN · Paper 01

Further practice · Attempt part c only.

Question 11 continued

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26. Question 11 · part c · 2 marks

2020 JAN · Paper 01

Further practice · Attempt part c only.

Question 11 continued

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(Total for Question 11 is 13 marks)

26. Question 11 · part c · 2 marks · Working and answer

Working and answer

OLevels.com SCHOOL

27. Question 5 · part c · 2 marks

2023 JUN · Paper 01

Further practice · Attempt part c only.

5 A solid cuboid has width x cm, length $4x$ cm and height h cm.
The volume of the cuboid is 75 cm^3 and the surface area of the cuboid is S cm^2

(a) Show that $S = 8x^2 + \frac{375}{2x}$

(4)

Given that x can vary, using calculus,

(b) (i) find to 3 significant figures, the value of x for which S is a minimum,

(ii) justify that this value of x gives a minimum value of S

(5)

(c) Find, to 3 significant figures, the minimum value of S

(2)

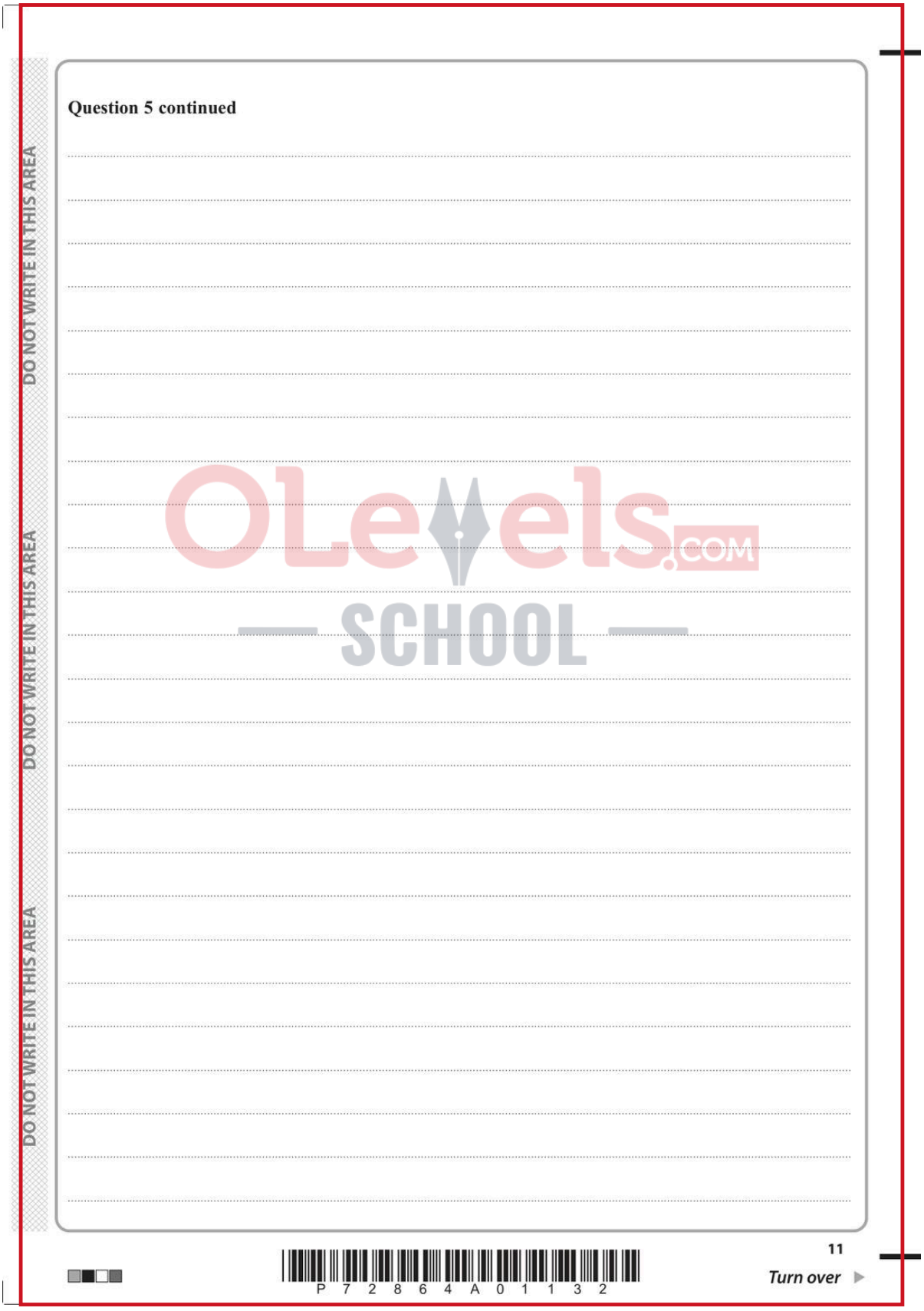
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27. Question 5 · part c · 2 marks

2023 JUN · Paper 01

Further practice · Attempt part c only.



27. Question 5 · part c · 2 marks

2023 JUN · Paper 01

Further practice · Attempt part c only.

Question 5 continued

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27. Question 5 · part c · 2 marks

2023 JUN · Paper 01

Further practice · Attempt part c only.

Question 5 continued

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(Total for Question 5 is 11 marks)

27. Question 5 · part c · 2 marks · Working and answer

Working and answer

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28. Question 8 · part c · 2 marks

2024 JUN · Paper 02

Further practice · Attempt part c only.

8

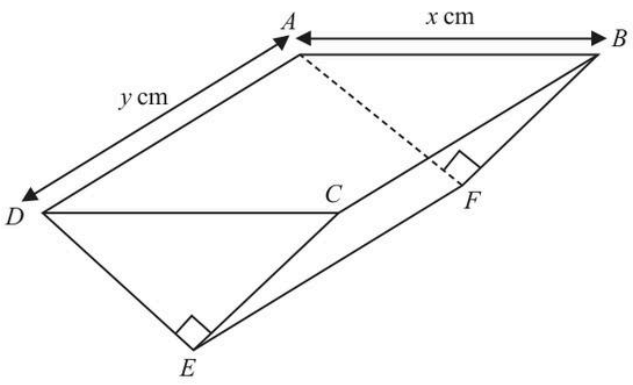


Diagram **NOT** accurately drawn

Figure 4

Figure 4 shows a solid right triangular prism $ABCDEF$

The cross section of the prism is an isosceles triangle.

- $\angle DEC = \angle AFB = 90°$
- $AB = DC = x$ cm
- $AD = BC = FE = y$ cm
- $AF = BF = DE = CE$

The triangular faces of the prism are vertical and the edges AD, BC and FE are horizontal.
The volume of the prism is 3.6 cm^3
The total external surface area of the prism is S cm^2

(a) Show that S satisfies the equation

$$S = \frac{x^2}{2} + \frac{72\left(\sqrt{2}+1\right)}{5x}$$

(4)

Given that x can vary,

(b) use calculus, to find to 3 significant figures, the value of x for which S is a minimum.
Justify that this value of x gives a minimum value of S

(4)

(c) Hence find, to 2 significant figures, the minimum value of S

(2)

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28. Question 8 · part c · 2 marks

2024 JUN · Paper 02

Further practice · Attempt part c only.

Question 8 continued

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28. Question 8 · part c · 2 marks

2024 JUN · Paper 02

Further practice · Attempt part c only.

Question 8 continued

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28. Question 8 · part c · 2 marks

2024 JUN · Paper 02

Further practice · Attempt part c only.

Question 8 continued

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(Total for Question 8 is 10 marks)

28. Question 8 · part c · 2 marks · Working and answer

Working and answer

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29. Question 5 · part c · 2 marks

2025 NOV · Paper 02

Further practice · Attempt part c only.

5

Diagram **NOT** accurately drawn

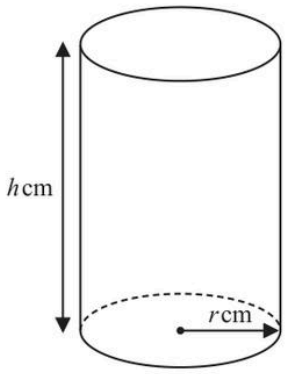


Figure 1

A solid right circular cylinder has base radius r cm and height h cm, as shown in Figure 1

The cylinder has a total surface area of 322π cm^2 and a volume of V cm^3

(a) Show that $V = \pi r\left(161 - r^2\right)$

(3)

Given that r can vary,

(b) use calculus to find, to 3 significant figures, the value of r for which V is a maximum, justifying that this value of r gives a maximum value of V

(5)

(c) Find, to 3 significant figures, the maximum value of V

(2)

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29. Question 5 · part c · 2 marks

2025 NOV · Paper 02

Further practice · Attempt part c only.

Question 5 continued

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(Total for Question 5 is 10 marks)

29. Question 5 · part c · 2 marks · Working and answer

Working and answer

OLevels.com SCHOOL

30. Question 11 · part c · 3 marks

2022 JUN · Paper 02 (regional R)

Further practice · Attempt part c only.

11

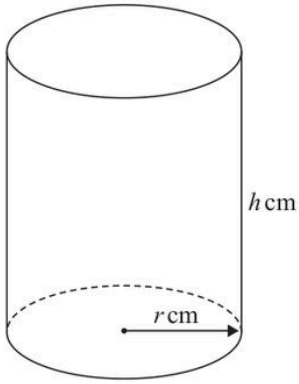


Diagram **NOT** accurately drawn

Figure 3

Figure 3 shows a solid metal right circular cylinder of radius r cm and height h cm.

The total surface area of the cylinder is $600\,\text{cm}^2$

The volume of the cylinder is $V\,\text{cm}^3$

(a) Show that $V = 300r - \pi r^3$ **(4)**

Given that r can vary,

(b) (i) use calculus to show that the exact value of r for which V is a maximum is

$$r = \sqrt{\frac{100}{\pi}}$$

(ii) justify that this value of r gives a maximum value of V **(5)**

The cylinder is melted down and reformed into a sphere of radius p cm.

(c) Find, to one decimal place, the greatest possible value of p **(3)**

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P 7 1 6 4 2 A 0 3 4 3 6

30. Question 11 · part c · 3 marks

2022 JUN · Paper 02 (regional R)

Further practice · Attempt part c only.

Question 11 continued

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30. Question 11 · part c · 3 marks

2022 JUN · Paper 02 (regional R)

Further practice · Attempt part c only.

Question 11 continued

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(Total for Question 11 is 12 marks)

30. Question 11 · part c · 3 marks · Working and answer

Working and answer

OLevels.com SCHOOL

31. Question 7 · part c · 2 marks

2024 NOV · Paper 01

Further practice · Attempt part c only.

7

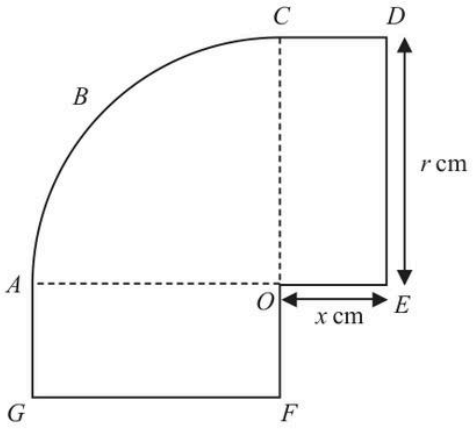


Diagram **NOT** accurately drawn

Figure 2

Figure 2 shows a shape $ABCDEOFG$

$ABCO$ is a quarter circle with radius r cm
$CDEO$ and $AOFG$ are congruent rectangles of length r cm and width x cm

The total area of the shape is $100\,\text{cm}^2$
The perimeter of the shape is P cm

(a) Show that $P = \dfrac{200}{r} + 2r$

(4)

(b) Use calculus to find the value of r for which P is a minimum, justifying that this value of r gives a minimum value of P

(5)

(c) Find the minimum value of P

(2)

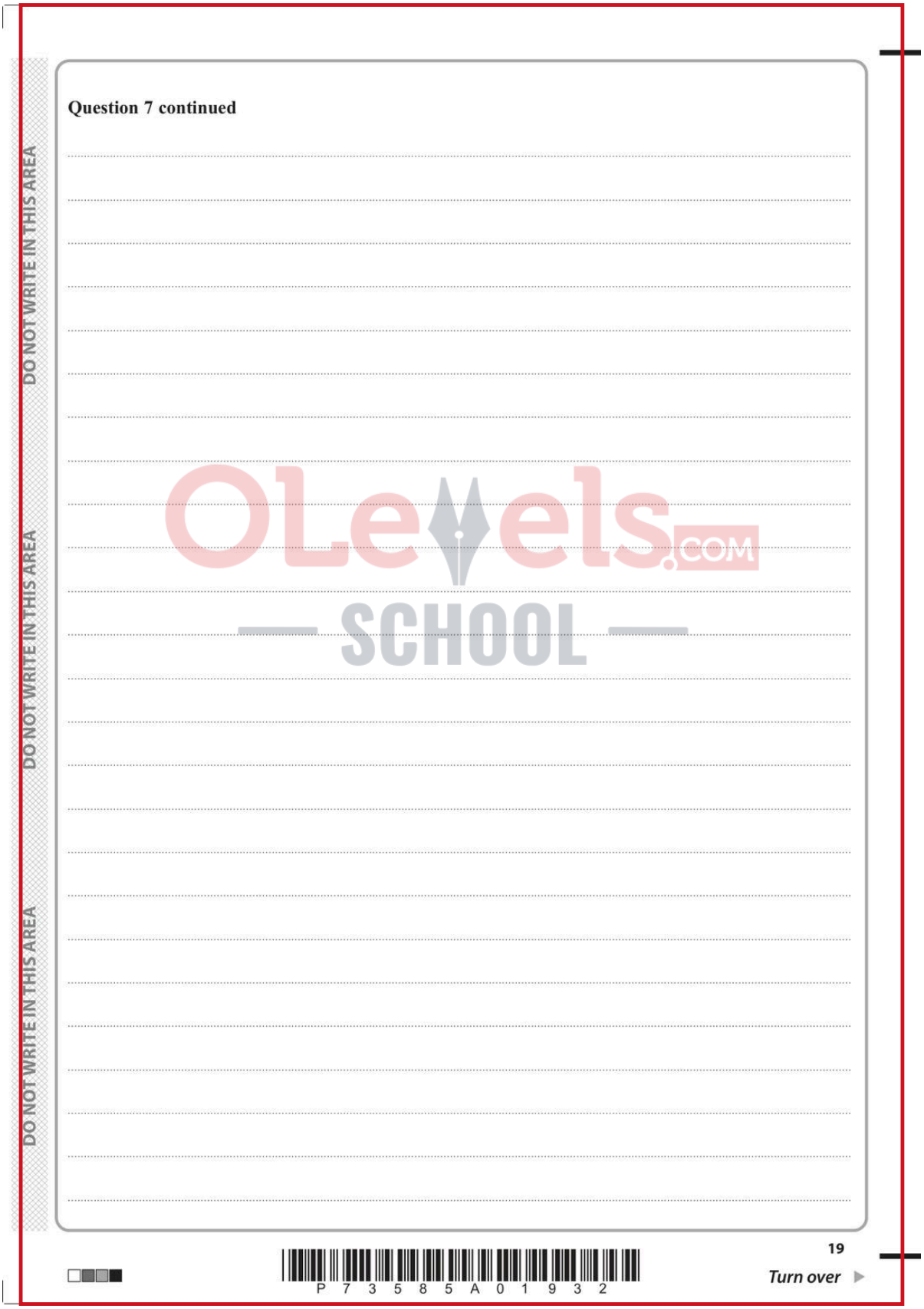
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31. Question 7 · part c · 2 marks

2024 NOV · Paper 01

Further practice · Attempt part c only.



Question 7 continued

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31. Question 7 · part c · 2 marks

2024 NOV · Paper 01

Further practice · Attempt part c only.

Question 7 continued

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31. Question 7 · part c · 2 marks

2024 NOV · Paper 01

Further practice · Attempt part c only.

Question 7 continued

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(Total for Question 7 is 11 marks)

31. Question 7 · part c · 2 marks · Working and answer

Working and answer

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32. Question 8 · part c · 2 marks

2022 JUN · Paper 02

Further practice · Attempt part c only.

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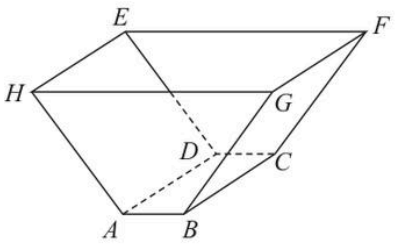


Diagram **NOT** accurately drawn

Figure 2

Figure 2 shows a waste paper basket in the shape of a right prism with 5 faces and a cross section that is a trapezium. The top, $EFGH$, of the waste paper basket is open.

The base of the prism $ABCD$ is a rectangle with

$$AB = DC = 2x\,\text{cm} \text{ and } AD = BC = h\,\text{cm}$$

The cross sections $HGBA$ and $EFCD$ are such that

$$EF = HG = 8x\,\text{cm} \text{ and } AH = BG = CF = DE = 5x\,\text{cm}$$

The top, $EFGH$, of the waste paper basket is such that

$$EH = FG = h\,\text{cm}$$

The volume of the waste paper basket is $2250\,\text{cm}^3$

The total surface area of the 5 faces of the waste paper basket is $S\,\text{cm}^2$

(a) Show that $S = 40x^2 + \dfrac{1350}{x}$

(5)

Given that x can vary,

(b) use calculus, to find, to 3 significant figures, the value of x for which S is a minimum.
Justify that this value of x gives a minimum value of S

(5)

(c) Find, to 3 significant figures, the minimum value of S

(2)

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32. Question 8 · part c · 2 marks

2022 JUN · Paper 02

Further practice · Attempt part c only.

Question 8 continued

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32. Question 8 · part c · 2 marks

2022 JUN · Paper 02

Further practice · Attempt part c only.

Question 8 continued

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32. Question 8 · part c · 2 marks

2022 JUN · Paper 02

Further practice · Attempt part c only.

Question 8 continued

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(Total for Question 8 is 12 marks)

32. Question 8 · part c · 2 marks · Working and answer

Working and answer

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33. Question 7 · part c · 2 marks

2022 JAN · Paper 01 (regional R)

Further practice · Attempt part c only.

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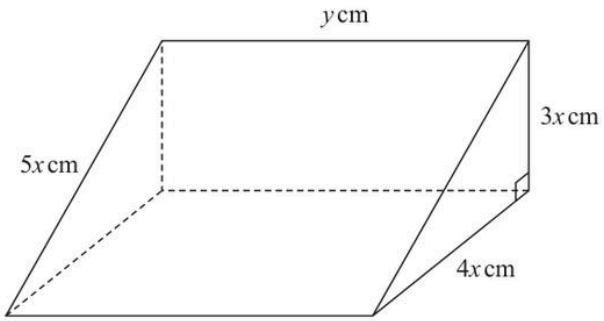


Diagram **NOT** accurately drawn

Figure 2

Figure 2 shows a block of wood in the shape of a right triangular prism.

The cross section of the prism is a right-angled triangle with sides of length $3x$ cm, $4x$ cm and $5x$ cm.

The length of the prism is y cm.

The total surface area of the five faces of the prism is 144 cm^2

The volume of the prism is V cm^3

(a) Show that

$$V = 72x - 6x^3$$

(5)

Given that x can vary,

(b) use calculus to find the value of x for which V is a maximum, justifying that this value gives a maximum value of V

(4)

(c) Find the maximum value of V

(2)

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33. Question 7 · part c · 2 marks

2022 JAN · Paper 01 (regional R)

Further practice · Attempt part c only.

Question 7 continued

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33. Question 7 · part c · 2 marks

2022 JAN · Paper 01 (regional R)

Further practice · Attempt part c only.

Question 7 continued

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33. Question 7 · part c · 2 marks

2022 JAN · Paper 01 (regional R)

Further practice · Attempt part c only.

Question 7 continued

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(Total for Question 7 is 11 marks)

33. Question 7 · part c · 2 marks · Working and answer

Working and answer

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Practice workbook · Rates of change with one chain-rule link

Linking a rate with respect to time to the rate of a length using one chain, such as dV/dt = dV/dh times dh/dt.

18 question parts: 1 worked-example retry and 17 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

OLevels.com SCHOOL

Type ID: FPM-CAL-14

1. Question 2 · part whole · 6 marks

2019 JUN · Paper 02

Worked-example retry · Attempt part whole only.

2 Oil is leaking from a pipe and forms a circular pool on a horizontal surface. The area of the surface of the pool is increasing at a constant rate of $8\,cm^2/s$. Find, in cm/s to 3 significant figures, the rate at which the radius of the pool is increasing when the area of the pool is $50\,cm^2$

(6)

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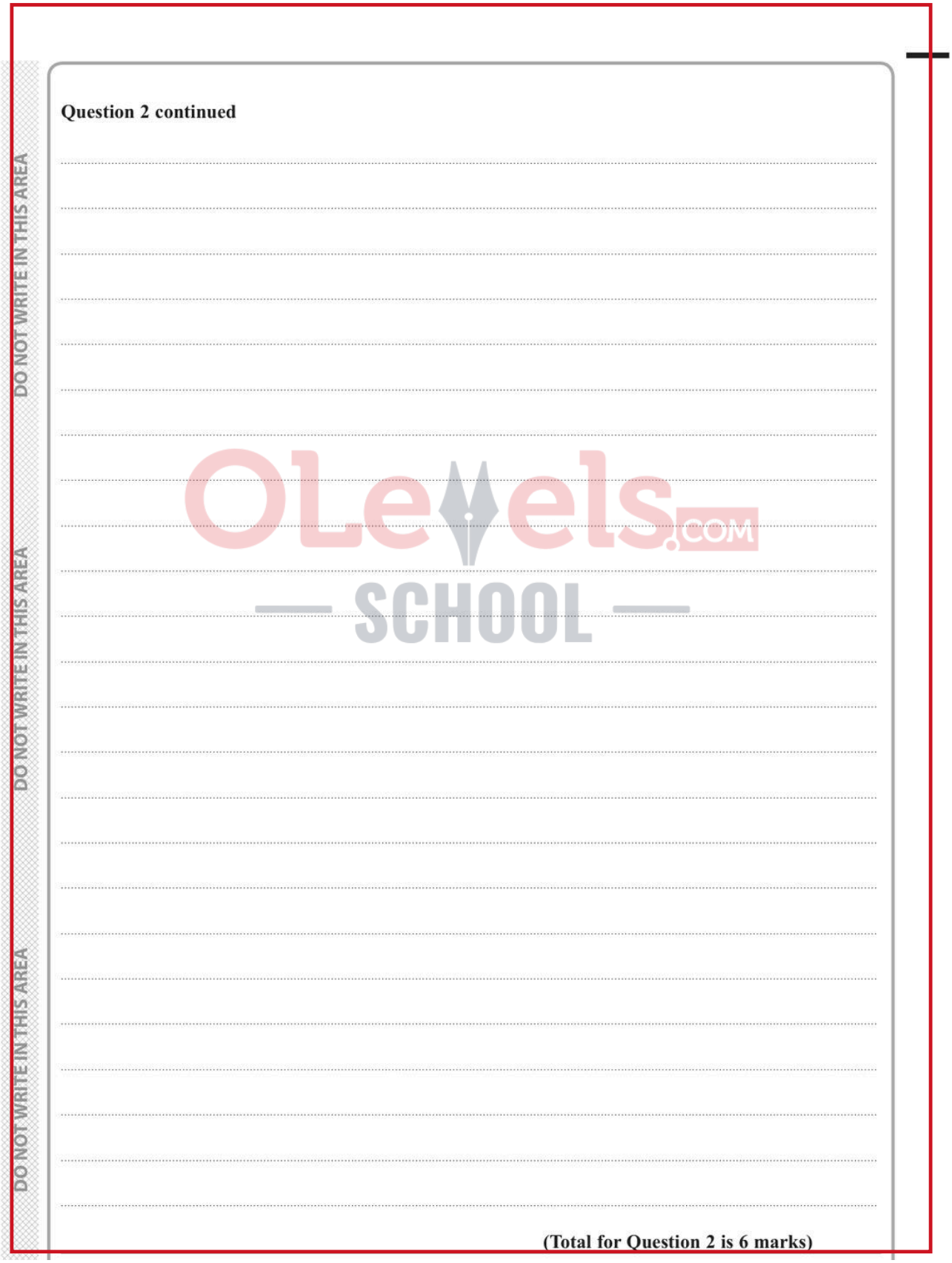
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1. Question 2 · part whole · 6 marks

2019 JUN · Paper 02

Worked-example retry · Attempt part whole only.



Question 2 continued

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(Total for Question 2 is 6 marks)

1. Question 2 · part whole · 6 marks · Working and answer

Working and answer

OLevels.com SCHOOL

2. Question 10 · part b · 3 marks

2020 NOV · Paper 01

Further practice · Attempt part b only.

10

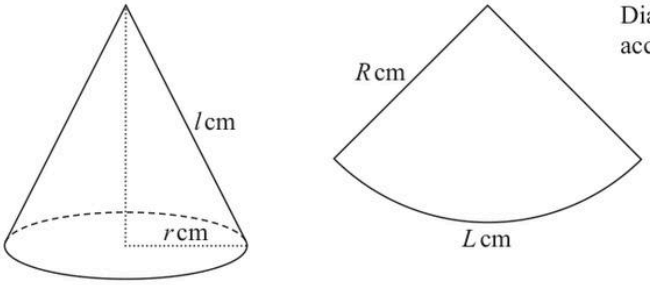


Diagram **NOT** accurately drawn

Figure 4

Figure 4 shows a right circular cone with base radius r cm and slant height l cm.
Figure 4 also shows a sector of a circle with radius R cm and arc length L cm.

The area of the curved surface of the cone is A cm^2

By considering how the sector of the circle can be folded to exactly form the curved surface of the cone with R and L suitably chosen,

(a) prove that $A = \pi rl$

(4)

Sand is poured onto a horizontal surface at a constant rate of 1.5 cm^3/s.
The sand forms a pile in the shape of a right circular cone with its base on the surface.
The curved surface area of the cone, A cm^2, increases in such a way that the height of the cone is always three times the radius of the base of the cone.

Given that $\frac{dA}{dr} = k\pi r$, where k is a constant,

(b) find the exact value of k.

(3)

(c) Calculate the rate, in cm^2/s, to 3 significant figures, at which the curved surface area of the pile is increasing when the height of the pile is 24 cm.

(5)

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2. Question 10 · part b · 3 marks · Working and answer

Working and answer

OLevels.com SCHOOL

3. Question 6 · part whole · 7 marks

2023 NOV · Paper 02

Further practice · Attempt part whole only.

6 The volume of oil in a container is V cm^3 when the height of the oil is h cm.
Oil is pouring into the container at a constant rate of 12 cm^3/s.
Given that $V = 3h^3$

find the exact rate, in cm/s, at which the height of the oil is increasing when $V = 1536$ cm^3

(7)

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P 7 3 5 8 6 A 0 1 6 3 2

3. Question 6 · part whole · 7 marks

2023 NOV · Paper 02

Further practice · Attempt part whole only.

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Question 6 continued

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(Total for Question 6 is 7 marks)

3. Question 6 · part whole · 7 marks · Working and answer

Working and answer

OLevels.com SCHOOL

4. Question 7 · part whole · 6 marks

2025 JUN · Paper 01

Further practice · Attempt part whole only.

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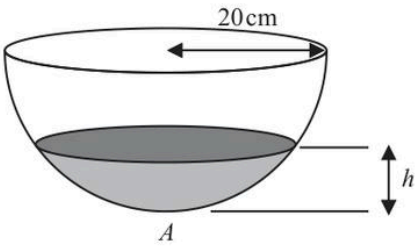


Diagram **NOT** accurately drawn

Figure 2

Figure 2 shows a hollow hemisphere with radius 20 cm

The hemisphere contains liquid, which is dripping out of a small hole at the lowest point A at a constant rate of k cm^3/s

At time t seconds after the liquid starts to drip from the hemisphere, the height of the liquid is h cm above A

The volume V cm^3 of liquid in the hemisphere is given by

$$V = \frac{\pi}{3}h^2(60 - h)$$

When $h = 12$, the height of the liquid is decreasing at a rate of $\frac{1}{60}$ cm/s

Find the value of k
Give your answer in terms of π

(6)

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4. Question 7 · part whole · 6 marks

2025 JUN · Paper 01

Further practice · Attempt part whole only.

Question 7 continued

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4. Question 7 · part whole · 6 marks

2025 JUN · Paper 01

Further practice · Attempt part whole only.

Question 7 continued

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4. Question 7 · part whole · 6 marks

2025 JUN · Paper 01

Further practice · Attempt part whole only.

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Question 7 continued

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(Total for Question 7 is 6 marks)

4. Question 7 · part whole · 6 marks · Working and answer

Working and answer

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5. Question 3 · part whole · 6 marks

2019 JUN · Paper 02 (regional R)

Further practice · Attempt part whole only.

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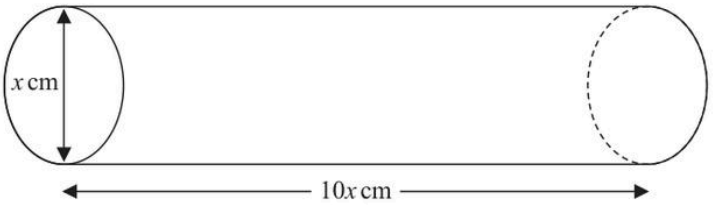


Diagram **NOT** accurately drawn

Figure 2

Figure 2 shows a solid right circular cylindrical metal rod.

The diameter of the rod is x cm and the length of the rod is $10x$ cm.

The rod is being heated so that the length of the rod is increasing at a rate of 0.005 cm/s.

Find the rate of increase, in cm^3/s to 2 significant figures, of the volume of the rod when $x = 3$

(6)

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5. Question 3 · part whole · 6 marks

2019 JUN · Paper 02 (regional R)

Further practice · Attempt part whole only.

Question 3 continued

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(Total for Question 3 is 6 marks)

5. Question 3 · part whole · 6 marks · Working and answer

Working and answer

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6. Question 8 · part a · 3 marks

2022 JAN · Paper 02 (regional R)

Further practice · Attempt part a only.

8 Liquid drips onto a large horizontal flat cloth, forming a circular stain.

The liquid starts to drip onto the cloth at time $t = 0$

The area of the stain increases at a constant rate of $1.5\,\text{cm}^2/\text{s}$

(a) Find, in terms of π, the radius of the stain at time $t = 4$ seconds.

(3)

(b) Find, in cm/s to 3 significant figures, the rate at which the radius of the stain is increasing at time $t = 4$ seconds.

(4)

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20

6. Question 8 · part a · 3 marks · Working and answer

Working and answer

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7. Question 8 · part b · 4 marks

2022 JAN · Paper 02 (regional R)

Further practice · Attempt part b only.

8 Liquid drips onto a large horizontal flat cloth, forming a circular stain.

The liquid starts to drip onto the cloth at time $t = 0$

The area of the stain increases at a constant rate of $1.5\,\text{cm}^2/\text{s}$

(a) Find, in terms of π, the radius of the stain at time $t = 4$ seconds.

(3)

(b) Find, in cm/s to 3 significant figures, the rate at which the radius of the stain is increasing at time $t = 4$ seconds.

(4)

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7. Question 8 · part b · 4 marks

2022 JAN · Paper 02 (regional R)

Further practice · Attempt part b only.

Question 8 continued

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(Total for Question 8 is 7 marks)

7. Question 8 · part b · 4 marks · Working and answer

Working and answer

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8. Question 6 · part whole · 6 marks

2023 JAN · Paper 02 (regional R)

Further practice · Attempt part whole only.

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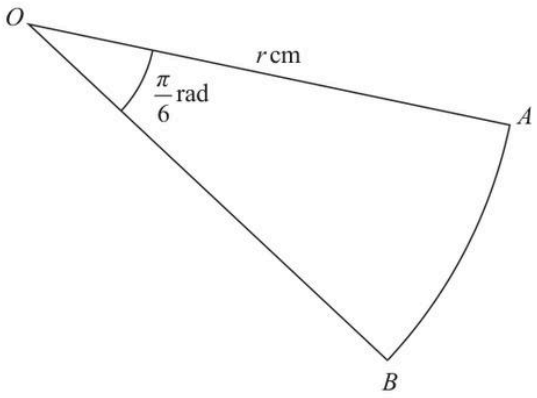


Diagram **NOT** accurately drawn

Figure 2

Figure 2 shows the sector OAB of a circle with centre O and radius r cm.

$$\angle AOB = \frac{\pi}{6} \text{ radians} \qquad OA = OB = r \text{ cm}$$

The area of the sector is increasing in such a way that the size of $\angle AOB$ remains constant, and the lengths OA and OB are both increasing at a constant rate of 0.2 cm/s

Find the exact rate of change, in cm^2/s, of the area of the sector when the length of arc AB is $\frac{5\pi}{2}$

(6)

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8. Question 6 · part whole · 6 marks

2023 JAN · Paper 02 (regional R)

Further practice · Attempt part whole only.

Question 6 continued

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(Total for Question 6 is 6 marks)

8. Question 6 · part whole · 6 marks · Working and answer

Working and answer

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9. Question 4 · part whole · 8 marks

2024 JUN · Paper 01

Further practice · Attempt part whole only.

4 The surface area of a sphere with radius r cm is increasing at a constant rate of 50π cm^2/s

Find, in cm^3, the exact volume of the sphere at the instant when the rate of increase of r is $\frac{5}{12}$ cm/s

(8)

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9. Question 4 · part whole · 8 marks

2024 JUN · Paper 01

Further practice · Attempt part whole only.

Question 4 continued

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(Total for Question 4 is 8 marks)

9. Question 4 · part whole · 8 marks · Working and answer

Working and answer

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10. Question 5 · part whole · 6 marks

2024 JUN · Paper 02 (regional R)

Further practice · Attempt part whole only.

5 The force F newtons between two magnetic poles is given by the formula

$$F = \frac{3}{20r^2}$$

where r is the distance, in centimetres, between the poles.

The distance between the two poles is increasing at a constant rate of 0.7 cm/s

Find the rate of change of F, in newtons/s to 3 significant figures, when the distance between the poles is 2.8 cm

(6)

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10. Question 5 · part whole · 6 marks

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Further practice · Attempt part whole only.

Question 5 continued

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(Total for Question 5 is 6 marks)

10. Question 5 · part whole · 6 marks · Working and answer

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11. Question 9 · part a · 5 marks

2021 JUN · Paper 02

Further practice · Attempt part a only.

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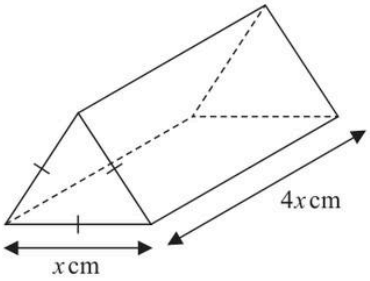


Diagram **NOT** accurately drawn

Figure 3

Figure 3 shows a metal solid S.

The solid is a right triangular prism.

The cross section of S is an equilateral triangle with sides of length x cm.
The length of S is $4x$ cm.

The prism is being heated so that the cross sectional area is increasing at a constant rate of $0.03\,\text{cm}^2/\text{s}$.

(a) Find, giving your answer to 3 significant figures, $\frac{dx}{dt}$ when $x = 2$ **(5)**

(b) Find the rate of increase, in cm^3/s, of the volume of S when $x = 2$ **(3)**

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11. Question 9 · part a · 5 marks · Working and answer

Working and answer

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12. Question 6 · part b · 4 marks

2020 JAN · Paper 02 (regional R)

Further practice · Attempt part b only.

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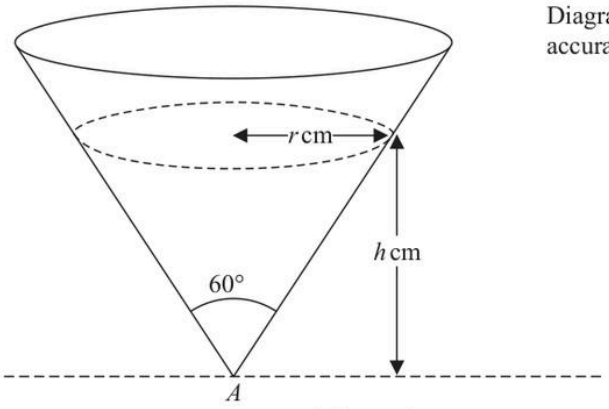


Diagram **NOT** accurately drawn

Figure 4

Figure 4 shows a hollow right circular cone fixed with its axis of symmetry vertical.

The cone is inverted and contains liquid, which is dripping out of a small hole at the vertex A of the cone at a constant rate of $0.9\,\text{cm}^3/\text{s}$.

At time t seconds after the liquid starts to drip from the cone, the height of the liquid is h cm above A. The volume of liquid in the cone at time t seconds is $V\,\text{cm}^3$

The vertical angle of the cone is $60°$

(a) Show that $V = \frac{1}{9}\pi h^3$

(2)

(b) Find, in cm/s to 3 significant figures, the rate at which the height of the liquid is decreasing when the height of the liquid in the cone above the vertex is 1.2 cm.

(4)

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12. Question 6 · part b · 4 marks

2020 JAN · Paper 02 (regional R)

Further practice · Attempt part b only.

Question 6 continued

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12. Question 6 · part b · 4 marks

2020 JAN · Paper 02 (regional R)

Further practice · Attempt part b only.

Question 6 continued

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12. Question 6 · part b · 4 marks

2020 JAN · Paper 02 (regional R)

Further practice · Attempt part b only.



Question 6 continued

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(Total for Question 6 is 6 marks)

12. Question 6 · part b · 4 marks · Working and answer

Working and answer

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13. Question 6 · part whole · 6 marks

2021 NOV · Paper 01

Further practice · Attempt part whole only.

6 The volume of a sphere with radius r cm is increasing at a constant rate of $3\,\text{cm}^3/\text{s}$.

Find the rate, in cm^2/s, at which the surface area of the sphere is increasing when $r = 10$

(6)

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13. Question 6 · part whole · 6 marks

2021 NOV · Paper 01

Further practice · Attempt part whole only.

Question 6 continued

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(Total for Question 6 is 6 marks)

13. Question 6 · part whole · 6 marks · Working and answer

Working and answer

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14. Question 2 · part whole · 6 marks

2022 JUN · Paper 02 (regional R)

Further practice · Attempt part whole only.

2 When poured from a pipe, concrete is formed into the shape of a cuboid with a square base of side x and with a height of $3x$

The volume of the cuboid increases at a constant rate of $8\,m^3/s$

Find the rate of increase, in m/s, of x when $x = 2$ metres.

(6)

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14. Question 2 · part whole · 6 marks

2022 JUN · Paper 02 (regional R)

Further practice · Attempt part whole only.

Question 2 continued

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(Total for Question 2 is 6 marks)

14. Question 2 · part whole · 6 marks · Working and answer

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15. Question 6 · part b · 5 marks

2023 JUN · Paper 02

Further practice · Attempt part b only.

6

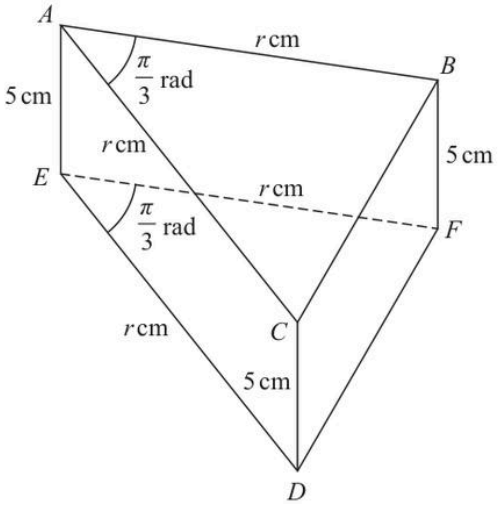


Diagram **NOT** accurately drawn

Figure 3

Figure 3 shows a right triangular prism $ABCDEF$. A cross section ABC of the prism is a triangle in which $AB = AC = r$ cm and $\angle CAB = \frac{\pi}{3}$ radians.

In the prism

$$AE = BF = CD = 5\,\text{cm} \qquad ED = EF = r\,\text{cm} \text{ and } \angle DEF = \frac{\pi}{3} \text{ radians}$$

(a) Show that the volume of the prism is $\frac{5\sqrt{3}}{4} r^2\ \text{cm}^3$

(1)

The volume of the prism is increasing in such a way that the size of $\angle CAB$ and the size of $\angle DEF$ remain constant and the length of AE, the length of BF and the length of CD remain constant.
The lengths of AB, AC, ED and EF are each increasing at a constant rate of $0.2\,\text{cm/s}$

(b) Find the exact rate of increase, in cm^3/s, of the volume of the prism when the area of the rectangular face $BCDF$ is $60\,\text{cm}^2$

(5)

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15. Question 6 · part b · 5 marks

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Further practice · Attempt part b only.

Question 6 continued

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15. Question 6 · part b · 5 marks

2023 JUN · Paper 02

Further practice · Attempt part b only.

Question 6 continued

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15. Question 6 · part b · 5 marks

2023 JUN · Paper 02

Further practice · Attempt part b only.

Question 6 continued

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(Total for Question 6 is 6 marks)

15. Question 6 · part b · 5 marks · Working and answer

Working and answer

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16. Question 5 · part whole · 5 marks

2024 NOV · Paper 02

Further practice · Attempt part whole only.

5 The height of liquid in a vessel P is h
The volume, V, of the liquid in P is given by $V = 6h^3$
Liquid is leaking from P at a constant rate of $36\,\text{cm}^3/\text{s}$

Find the exact rate of change, in cm/s, of h when $V = 384\ \text{cm}^3$

(5)

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16. Question 5 · part whole · 5 marks

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Further practice · Attempt part whole only.

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Question 5 continued

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(Total for Question 5 is 5 marks)

16. Question 5 · part whole · 5 marks · Working and answer

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OLevels.com SCHOOL

17. Question 7 · part a · 4 marks

2020 NOV · Paper 01 (regional R)

Further practice · Attempt part a only.

7 The length of each side of a cube S_1 is increasing at a constant rate of 0.1 m/s.

(a) Find, in m^3/s, the rate of increase of the volume of the cube S_1 when the length of each side of the cube is 2 m.

(4)

The total surface area of a different cube S_2 is increasing at a constant rate of 0.05 m^2/s.

(b) Find in m^3/s, the rate of increase of the volume of the cube S_2 when the length of each side of the cube is 6 m.

(5)

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17. Question 7 · part a · 4 marks · Working and answer

Working and answer

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18. Question 11 · part whole · 8 marks

2025 JUN · Paper 01 (regional R)

Further practice · Attempt part whole only.

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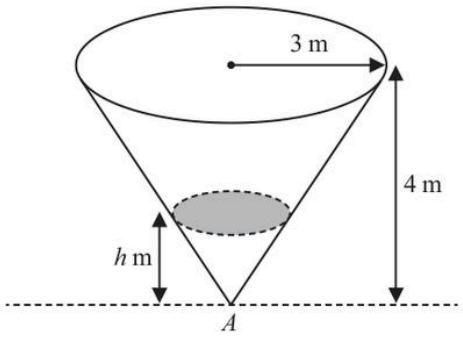


Diagram **NOT** accurately drawn

Figure 3

Figure 3 shows a hollow right circular cone with radius 3 metres and height 4 metres above the vertex A

The cone is fixed with its axis of symmetry vertical.
The cone is initially empty.

Water pours into the cone at a constant rate of 9π m^3/s
At time t seconds after the water starts to pour into the cone, the height of the water is h metres above A

Find, in m/s, the rate at which the height of the water is increasing at the instant when $h = 2$

(8)

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18. Question 11 · part whole · 8 marks

2025 JUN · Paper 01 (regional R)

Further practice · Attempt part whole only.

Question 11 continued

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18. Question 11 · part whole · 8 marks

2025 JUN · Paper 01 (regional R)

Further practice · Attempt part whole only.

Question 11 continued

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(Total for Question 11 is 8 marks)

18. Question 11 · part whole · 8 marks · Working and answer

Working and answer

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